Obstetric Anaesthesia Guidelines/Handbook

Summary

This document contains information and guidance relating to the safe delivery of obstetric anaesthesia and analgesia. It should be read by all who provide obstetric anaesthesia. Anaesthetists new to this Trust should recognise that practise may differ from maternity units they have previously worked in and should follow the guidelines within this document.

While guidelines should be followed wherever possible, it is recognised that in some situations individualised care may be required. This should be provided in the light of the clinical situation and with senior consultation where necessary.

Guidelines for some specific obstetric conditions (e.g. pre-eclampsia, sepsis etc) are outside the scope of this handbook but are available as separate documents on the Trust intranet and should be consulted and followed where necessary.

Obstetric Anaesthetic Consultants

The following consultants provide obstetric anaesthesia cover:

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Record keeping

Obstetric care is at times challenging and complications and problems do occur. Clear, contemporaneous and thorough record keeping is essential at all times and is especially necessary should problems or potential problems arise.

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1 Main Duties

Organisation

The roles of the obstetric anaesthetist extend beyond provision of obstetric anaesthesia and analgesia and include; assessment of low and high-risk women, management of obstetric emergencies and provision of maternal critical care.

The obstetric anaesthetist is an integral member of a multi-professional team. Close working relationships exist between these team members in Exeter and this is key to ensuring on-going delivery of safe and high quality care.

The shift times, including time for handover are:

Days: 08.00 – 20.15
Nights: 20.00 – 08.15

There is a dedicated pager (#981) that must be carried at all times and handed over to the incoming anaesthetist at changeover of shifts.

Labour ward hand-over meetings:

Multi-disciplinary discussion of women in labour and other planned and potential activities within the maternity unit take place in the labour ward office at 08.00 and 20.00.

Attendance is mandatory unless busy in theatre or on the labour ward.
The obstetric anaesthetist finishing their shift must attend the 08.00 meeting.

The incoming obstetric anaesthetist should write their name on the white board in the Labour Ward office, sign the appropriate page in the hand-over file in the labour ward office and ensure the pager is working.

Any high risk women on labour ward should be highlighted at the morning meeting and seen jointly by the anaesthetic and obstetric team.

Anaesthetic hand-over

Hand-over from outgoing to incoming anaesthetist should detail the following:

1. **S**ick woman on labour ward, ante or post-natal ward or elsewhere in the hospital.
2. **A**t risk women – e.g. women with potential or on-going problems where anaesthetic help may be required, e.g. HDU patients, pain issues, difficult IV access.
3. **F**ollow ups – women that require follow-up (either in hospital or by telephone) as a result of an anaesthetic complication e.g. post-dural puncture headache, neurological problem etc.
4. **E**pidurals - women with on-going epidural analgesia on the labour ward should be handed over along with any issues regarding analgesia.

The acronym ***SAFE*** can be used to enhance the quality of this handover.

Team Briefing

Following the 08.00 handover meeting in the labour ward office, the obstetric and anaesthetic team must meet each day with theatre and ODP staff in the anaesthetic room of Theatre 1 for a team brief.

Daily responsibilities

Daily responsibilities of the obstetric anaesthetist include:

- Checking the anaesthetic machine and emergency drugs
- Reviewing women on labour ward with ongoing epidural analgesia
- Provision of regional and general anaesthesia for obstetric procedures
- Provision of epidural analgesia for women in labour
- Antenatal and postnatal care and review of women having anaesthetic interventions
- Review of all women who have had an anaesthetic intervention the previous day
- Follow-up patients should be accessed on the An Obs Telephone Report on MyCare. These can be added retrospectively if required (refer to the Epic tip sheets in the Appendix - telephone follow up section). Difficult cases can be flagged up on the "handover board" in the obstetric anaesthetic office.
- Assistance with the management of women with medical or obstetric complications
- Assistance at obstetric emergencies, including maternal collapse, haemorrhage and eclampsia
- Anaesthesia for minor gynaecological emergencies

From midnight to 0800 the obstetric anaesthetist carries the 1st on-call anaesthetic pager (#485). The departing 1st on-call anaesthetist will give the pager to the obstetric anaesthetist at 2400 and the incoming 1st on-call anaesthetist will collect the pager at 0800 the following morning. There is a 4th On-Call Anaesthetist to support the obstetric anaesthetist. They should be called for advice and help (refer to the escalation policy in the Appendix).

Work outside the maternity unit by the obstetric anaesthetist

'Out-of-hours' the obstetric anaesthetist may be asked to provide assistance in other areas of the hospital including; the Emergency Department, main theatres, ITU and wards. It is imperative the obstetric anaesthetist is able to leave this work to attend an obstetric emergency should one occur – e.g. category 1 caesarean section, major haemorrhage etc, and should not take on duties which cannot be safely handed over.

If patients require emergency surgery in main theatres and the consultant anaesthetist/ 4th On Call not immediately available, the obstetric anaesthetist, if not busy on the labour ward, should review and assess these patients to enable their timely transfer to theatre and assist in theatre if necessary.

The obstetric anaesthetist must be immediately available for obstetric emergencies at all times

Seeking Assistance

As a general rule, if a trainee is unfamiliar with a clinical problem, is faced with more than one emergency at once or has a case they are unhappy to undertake, they should seek the assistance of a senior colleague, 4th On Call in the first instance and then the General Consultant Anaesthetist on call. **If in doubt, seek assistance. Refer to the obstetric escaltation policy in the Appendix.**

Record keeping

For obstetric patients, the following should be completed:.

Women considered at high risk, or with specific issues that may make anaesthesia more challenging, will often have been seen by one of the obstetric anaesthetic consultants during their pregnancy in the High Risk Pre-assessment Clinic.

Documentation in Obstetric Anaesthesia

Approximately 25% of all anaesthetic related closed litigation claims are related to regional anaesthesia in obstetrics. Comprehensive documentation is not only a clinical requirement, but also important from a medico-legal perspective.

In addition to standard documentation as recommended by the RCoA/AAGBI, the following should be recorded when completing an anaesthetic chart for an obstetric anaesthesia specific risks for regional and general anaesthesia should be discussed with the woman (detailed later).

References

Cook TM, Bland L, Mihai R, Scott S. Litigation related to anaesthesia: an analysis of claims against the NHS in England 1995–2007. Anaesthesia 2009; 64: 706–18

2 Preoperative Assessment and Preparation

All women undergoing any anaesthetic intervention should be assessed pre-operatively including a targeted history and airway examination. Anaesthetic options should be explained, including their advantages and disadvantages.

Women booked for elective caesarean section are seen in the Fetal and Maternal Assessment Unit (FMAU) the day before surgery (from 11.00). Staff in FMAU will page the obstetric anaesthetist when the women arrive - please attend promptly to limit the time they spend waiting in FMAU. If you are unable to attend, let the consultant know that these women are waiting for anaesthetic review.

For procedures under general anaesthesia the following potential complications should be discussed:

- Muscle pain – 1 in 3
- Damage to teeth – 1 in 4500
- Potential difficulties with intubation - Intubation failure 1 in 250
- Risk of gastric acid regurgitation and aspiration – 1 in 1000
- Risk of awareness – 1 in 400
- Severe allergic reaction – 1 in 10000
- Neonatal sedation
- Post-operative pain

For procedures under regional anaesthesia the following potential complications should be covered: (See OAA information card – Appendix 1)

- Significant drop in BP and associated nausea: 1 in 5
- Itching
- Intraoperative sensation, including pain
- Conversion to GA
- Severe headache – 1 in 100 (epidural), 1 in 500 (spinal)
- Nerve damage, which may be temporary or permanent:
  Temporary 1 in 1000, Permanent 1 in 13 000
- Accidental unconsciousness – 1 in 100 000

Consent should be obtained for the use of rectal analgesia when appropriate.

If the case is urgent, pre-operative assessment and discussion may have to be done quickly. On such occasions the anaesthetist should make all reasonable efforts to ensure that patient safety is maintained.

Please inform the patient that we will see them post-operatively on the ward. If they go home before being seen then to expect a follow-up phone call.

High Risk Anaesthetic pre-assessment Clinic

A high risk clinic is run once a month where we see referrals regarding women who are predicted to be more complicated than normal so that an anaesthetic management plan can be made. A high risk anaesthetic clinic letter should be available on Mycare. Some high risk patients will have been seen by the Bristol maternal cardiac service (see separate maternity cardiac guideline). High risk plans

may be printed and attached to the cork board in the obstetric anaesthetic office and also uploaded into the Media section of MyCare.

In addition to these women we would like to see post delivery those women who had a blood patch as a treatment for a post dural puncture headache and all those with a neurological deficit thought to be secondary to regional anaesthesia. For those with a neurological injury a delay of 3 months from injury is required to that any improvement can be documented.

An appointment can be booked via an Order through MyCare. This has to be done as in an outpatient setting on MyCare. If you search the patient through the “patient station” this will allow an order for High risk obs anaesthetic clinic. You will need to specify when this patient should be seen.

Premedication

- Avoid sedative premedication wherever possible.
- Oral omeprazole for elective cases, 20mg at 1800hrs the night before surgery and 0600hrs the morning of surgery. For emergency cases, oral omeprazole should be given as soon as the decision for surgery is made. In cases where time is limited and general anaesthesia is urgently required, do not delay surgery.
- Consider 30ml of 0.3 molar sodium citrate to women about to receive a general anaesthetic and in women receiving a regional anaesthetic for a non-elective caesarean section.

Blood grouping/Cross-matching

Blood does not need to be cross-matched for caesarean section unless specific risk factors are present, e.g. placenta accreta, atypical antibodies.

All women requiring a caesarean section should be ‘group and saved’ (2 samples are required). For elective caesareans, this is performed, along with a FBC, in FMAU the day before elective surgery.

Most women whose maternity care has been managed in Exeter throughout their pregnancy will have had a previous ‘group and save’ taken at 12 weeks, so just one further sample will be required for both elective and emergency cases.

Beware however that some women, especially during the holiday season will have had their care in another hospital and may require 2 blood samples for grouping should they present as an emergency.

Two units of O-neg blood are kept in the Labour Ward blood fridge, which should be used in an emergency if a transfusion is required before cross-matched blood is available.

3 Caesarean Section: Definitions of urgency and standard procedures

Whether caesarean section is performed under general or regional anaesthesia, the following measures must be taken in all cases:

- Check all equipment and drugs which may be needed, including the various aids for difficult intubation (the difficult intubation trolley is outside theatre 1)
- Ensure antacid prophylaxis has been given
- Secure IV access – at least 16G
- Left lateral tilt until delivery
- Give a uterotonic at delivery, either:

  - **Carbetocin 100mcg** IV diluted in 10ml saline and given by slow injection over 1 minute after delivery and clamping of the cord

Or

  - **Syntocinon 5 units** IV diluted in 10ml saline and given by slow injection over 1 minute after delivery and clamping of the cord, followed by **syntocinon infusion of 30 units syntocinon in 500ml saline over 2- 4 hours** by infusion pump (125ml/h)

Which uterotonic drug to give should be discussed at 'Time Out' prior to surgery.

Classification of Caesarean Section Urgency

Classification of the urgency of caesarean section may:

- Aid communication relating to urgency of delivery
- Facilitate local and national data collection
- Facilitate retrospective audit of outcomes

Category 1: Immediate threat to the life of the woman or fetus

For example;

- Severe fetal bradycardia
- Severe abnormal CTG
- Cord prolapse with fetal compromise
- Uterine rupture
- Massive placental abruption
- Fetal pH<7.2
- Maternal cardiac arrest

Category 2: Maternal or fetal compromise which is not immediately life-threatening

For example;

- Failure to progress associated with concerning CTG

Category 3: No maternal or fetal compromise, but early delivery required

For example**;**

- Malpresentation in labour (e.g. undiagnosed breech)
- Planned elective caesarean section presenting in labour
- Pre-eclampsia / eclampsia needing caesarean
- Other serious medical conditions

Category 4: Delivery timed to suit the woman or staff

For example;

- **Elective caesarean section**

Decision-to-delivery intervals

Current NICE guidance states:

"Delivery at CS for maternal or fetal compromise should be accomplished as quickly as possible, **taking into account that rapid delivery has the potential to do harm**. A decision-to-delivery interval of less than 30 minutes is not in itself critical in influencing baby outcome, but has been an accepted audit standard for response to emergencies within maternity services."

Actions for when a category 1 caesarean section is required

All staff involved in obstetric emergencies carry a pager that will sound should a category 1 caesarean or other obstetric emergency occur.

To declare a category 1 caesarean, dial 2222 and say "Category 1 emergency caesarean section labour ward".

The on-call anaesthetist should proceed directly to labour ward.

For all category 1 caesarean sections:

Transfer to theatre immediately

- Do not waste time by applying TED stockings in the room or changing into a gown. Ensure women is placed and transferred to theatre on her side.

In theatre:

- Maintain the woman in the left lateral position and perform **Intrauterine Resuscitation – SPOILT** if appropriate: see below)
- Consider if regional anaesthesia acceptable (epidural top-up, or single shot spinal)
- Continue CTG monitoring while preparing for anaesthesia
- Consider Sodium Citrate 30ml

In some situations, general anaesthesia may be the most appropriate option but there are many circumstances when top-up of an in-situ epidural or insertion of a spinal block are appropriate.

The urgency of the case and the suitability of regional or general anaesthesia should be clarified with the obstetric team.

If there are serious concerns with providing general anaesthesia safely (e.g. severe obesity, potentially difficult airway), then general anaesthesia **should not be** performed simply because of the time pressures associated with a category 1 caesarean. In such cases, help should be sought early and attempts at regional anaesthesia commenced. Clear communication with the obstetric, midwifery and patient should be observed closely.

Spinal anaesthesia for a category 1 caesarean

- Continuous CTG monitoring
- Insert spinal with woman in the **left lateral position** (note: in CWH theatre 1, this will mean the woman will face the opposite way as she would when having a spinal in the sitting position)
- The left lateral positions limits aorto-caval compression and facilitates continuous monitoring of the fetus
- Adding diamorphine to the spinal mix is not necessary (preparation takes time and errors may occur if prepared in haste). Instead, add fentanyl 15-20 mcg.
- Aim to establish the usual standard of sensory block. If the block is ascending rapidly but has not yet reached the defined dermatomal level, discuss with the woman the importance of prompt surgery and that although she is likely to be comfortable there is a chance that she may feel pain and if so, general anaesthesia can be provided. Document this discussion and her consent to continue if applicable.

If, a decision is made to provide regional anaesthesia for a category 1 caesarean but it proves difficult to establish an adequate block, then the anaesthetist must not persist but convert to general anaesthesia unless there are serious risks associated with this technique.

Antibiotic prophylaxis

Standard antibiotic choice if for Augmentin 1.2g before KTS in caesarean section and instrumental deliveries. Antibiotic guidelines indicate what to do if the patient has a mild/severe penicillin allergy, and also for MRSA positive patients.

Further doses of antibiotic must be given in cases where there has been more than 1500ml blood loss.

See Appendix for the Obstetric prophylactic antibiotic table.

4 Intra-uterine Resuscitation

As soon as a category 1 caesarean section is declared for fetal indications intra-uterine resuscitation can be performed.
Instituting these simple steps may enable regional anaesthesia to be provided instead of proceeding straight to delivery under general anaesthesia, which may be associated with a worse maternal and fetal outcome.

The relevant interventions involved in intrauterine resuscitation may be remembered by the acronym **SPOILT**:

- **S**yntocinon off
- **P**osition – turn to full left lateral position
- **O**xygen – Provide high flow O_2 via Hudson mask
- **I**ntravenous fluids - commence or speed up Hartmann's solution
- **L**ow blood pressure? Consider giving dose of vasopressor
- **T**ocolysis – Not if obvious abruption or APH.
    - Terbutaline 0.25 mg subcutaneously or,
    - GTN sublingual spray 2 puffs repeated after 1 minute until contractions stop, maximum 3 doses

References:

Thurlow J, Kinsella M. Intrauterine resuscitation: active management of fetal distress. IJOA 2002; 11: 105

McDonnell NJ, Paech MJ. General anaesthesia for caesarean delivery: is the time saved worth the potential risks? Aust NZ J Obstet Gynaecol. 2012; 52: 311-2

5 Principles of Obstetric Regional Anaesthesia

The increased use of regional anaesthesia in obstetric over the last 20 years has resulted in a reduction in anaesthesia-related morbidity and mortality. However, regional anaesthesia at times can be challenging and is associated with a number of serious complications.

Preparation for regional anaesthesia (fasting, antacid prophylaxis, positioning etc) should be similar to preparation for general anaesthesia - **regional anaesthesia may fail or conversion to general anaesthesia may be necessary for other reasons.**

Factors that may make regional anaesthesia particularly challenging in pregnant women include:

- Requirement for a high sensory block (loss of ice sensation to T4 and loss of light touch sensation to T6) and consequent impact of sympathetic block and complications of high block
- Decreased dose requirement for spinal or epidural drugs
- Aortocaval compression
- The need to establish an adequate block in a short period of time in some circumstances (e.g. emergency caesarean section)

Principles to address the above include:

- Large bore IV access with crystalloid running
- Close monitoring of blood pressure with rapid correction of hypotension using vasopressor (see below)
- Left lateral tilt until delivery
- Close attention to assessment of sensory block

Contraindications to regional anaesthesia /analgesia

- Coagulopathy
- Low platelet count – see below
- Local infection at the site of insertion
- Severe sepsis
- Hypovolaemia
- Patient refusal
- Some intracranial pathology (e.g. raised intracranial pressure, major spina bifida, symptomatic brain tumour)

In addition, caution is advised in:

- Fixed cardiac output states (e.g. aortic stenosis, HOCM)
- Some neurological conditions e.g. spina bifida, multiple sclerosis etc where relapse of the CNS disease may be blamed on the neuraxial block

Thrombocytopaenia and disorders of coagulation

There is no specific platelet count above which regional anaesthesia or analgesia can be considered safe.

Causes of thrombocytopaenia in pregnancy include:

- Gestational thrombocytopaenia
- Pre-eclampsia (including HELLP)
- Autoimmune thrombocytopaenia
- Sepsis

The rate at which the platelet count is falling may be more important than the absolute count. Always review the previous platelet count to establish how dynamic the platelet decline is. **The platelet count can fall rapidly in pre-eclampsia and should be checked every 6 hours in severe cases.**

The following is taken from the 2013 AAGBI Guideline – Regional Anaesthesia and patients with abnormalities of coagulation.

- Risk is a continuum that runs from 'normal risk' to 'very high risk'
- An abnormality of coagulation however severe is always a *relative* contraindication to the use of a regional anaesthetic technique
- A platelet count of >75 x10^9/l has been proposed as an adequate level for regional blocks when there are no risk factors and the count is not decreasing
- In pre-eclampsia a decreasing platelet count is associated with other coagulation abnormalities and this is assumed to be the case once the platelet count decreases to below 100 x 10^9/l
- In idiopathic thrombocytopaenic purpura (ITP) and gestational thrombocytopaenia, there are reduced platelet numbers but normal function
- You can check the coagulation status when platelets 75-100 x10^9/l, by using TEG POC testing +/- sending a clotting screen.

Risk factor	Normal risk	Increased risk	High risk	Very high risk
ITP	Platelets >75 x10^9/l within 24h of block	Platelets 50-75 x 10^9/l	Platelets 20-75 x 10^9/l	Platelets <20 x 10^9/l
Pre-eclampsia	Platelets >100 x10^9/l within 6h of block	Platelets 75-100 x10^9/l (stable) *and* normal coagulation tests	Platelets 75-100 x10^9/l (decreasing) *and* normal coagulation tests	Platelets <75 x10^9/l or abnormal coagulation tests

- Women on low dose aspirin can undergo neuraxial blockade in the normal way up to 150mg.

Other considerations:

- Epidural placement may present more risk of neuraxial haematoma then a single-shot spinal due to increased risk of bleeding from epidural veins
- In some situations a general anaesthetic may expose the patient to greater risks than a regional technique in the presence of abnormal coagulation
- Apply the same cautions to removal of an epidural catheter in the presence of low platelets. Consider rechecking the platelets prior to epidural removal if the count was <100 x 10^9/l on insertion and > 6 hours since last blood test.
- **Experienced clinicians should be involved in decisions about whether or not to perform a regional technique in the presence of abnormal coagulation and the woman should be given all the information she needs to make an informed choice.**

6 Anaesthesia Techniques for Caesarean Section

Single shot spinal anaesthesia

- Usual sterile precautions as for epidural insertion
- Use an NRfit 25G pencilpoint (Whitacre) spinal needle via an introducer
- There is a risk of dural puncture if the introducer needle is inserted too far – especially likely in thin patients – be cautious
- Using a filter needle, prepare 2.0 – 2.5 ml hyperbaric 0.5% bupivacaine in a 3 ml NRfit syringe.
- Add 300mcg diamorphine to the bupivacaine. Prepare the diamorphine by diluting diamorphine 5mg in 5 ml 0.9% NaCl (to make a 1mg/ml solution) and add 300mcg (0.3ml) from this mix to the hyperbaric bupivacaine using a 1ml syringe
- **REMOVE THE SYRINGE OF DIAMORPHINE IN SOLUTION FROM THE TROLLEY TO AVOID INADVERTENT INTRATHECAL INJECTION**
- In emergency situations if time is limited, omit diamorphine and instead add 15 mcg (0.3ml) fentanyl
- Once the spinal drugs are administered, commence a phenylephrine infusion (see below) and lie the woman on her right side for a few minutes before placing supine with leftward table tilt to limit aorto-caval compression
- Commence BP monitoring every minute for the first few minutes while the spinal block develops
- Strongly consider the use of dexamethasone 3.3mg (as per PROSPECT trial). This has been shown to increase the length and quality of the block as well as reduce the incidence of nausea.
- In the event of diamorphine shortage. Morphine/ fentanyl substitute 100mcg preservative free morphine and 20 mcg fentanyl (see appendix for drawing up guidance). The nausea rate is greater than with diamorphine so strongly consider an anti-emetic e.g. dexamethasone/ ondansetron.

Combined Spinal-epidural (CSE) Anaesthesia

Indications for CSE for caesarean section include:

- Anticipated prolonged surgery that may outlast the original spinal block
  - E.g. morbid obesity, previous abdominal surgery, placenta praevia
- Women in which a normal spinal dose may lead to an unpredictable block height
  - E.g. short stature, multiple pregnancy, previous high block with single shot spinal
- Cases in which spinal-induced hypotension may be harmful, e.g. cardiac disease (aortic stenosis, HOCM). In these cases *'low-dose sequential CSE anaesthesia'* can be provided. This is a low-dose spinal anaesthesia followed by epidural top-up to more slowly achieve the desired block height.

CSE techniques

Needle-through-needle

Use a Portex CSEcure set (16G Tuohy with a long 26G pencilpoint spinal needle). The epidural space is identified with the Tuohy needle through which the long spinal

needle is passed into the intrathecal space and spinal drugs given. The spinal needle is then removed and the epidural catheter passed.
In this technique (spinal block first) the epidural catheter is not tested and care should be taken in administering the first epidural local anaesthetic dose.

Separate space

An epidural is performed in the usual way, the catheter inserted and tested with 3 ml bupivacaine 0.5% to exclude intrathecal placement. A spinal block is then performed at the level below the epidural site.

CSE local anaesthetic dosing

In cases that are likely to be prolonged, give a similar intrathecal dose to that used for a single shot spinal (see above). Consider using intrathecal fentanyl 15mcg and give epidural diamorphine 3mg at the end of the procedure.

If there are concerns over block height or risk of harm from hypotension (e.g. cardiac disease), smaller intrathecal doses can be used in the presence of an epidural. Doses as low as 7.5mg (1.5ml) hyperbaric bupivacaine may be effective with the back-up of an epidural top-up.

- A low-dose spinal reduces hypotension and nausea but has a reduced duration of action and is associated with a greater incidence of intraoperative pain.
- If using a low-dose of intrathecal bupivacaine a satisfactory block height may be achieved by injection of a volume of saline (e.g.8ml) via the epidural catheter on completion of the CSE (Epidural Volume Extension (EVE) or by incremental doses of epidural local anaesthetic (e.g. 5ml boluses of levobupivacaine 0.5%) (low-dose sequential CSE).
- If surgery is prolonged and a CSE is sited, top-up the epidural before pain occurs (~ 45 minutes after CSE sited if uterus is still not closed).

  At the end of surgery the epidural catheter should be removed by the anaesthetist and documented on MyCare (ensure catheter tip intact).
- Prescribe the first dose of dalteparin (Fragmin) for 4 hours **after removal of** the epidural catheter.

Epidural Top-up

If an epidural for labour pain is in-situ, it provides an effective means to establish regional anaesthesia for caesarean section. However, an epidural that has performed poorly for labour pain relief is more likely to provide inadequate anaesthesia if topped-up.

Avoid topping up an inadequate epidural and instead remove and administer a spinal anaesthetic.

Administering a spinal anaesthetic *after* the epidural has been topped-up may result in a high sensory block.

The ideal characteristics of an epidural solution for top-up for caesarean section include:

- Provision of a rapid onset of block
- Provision of reliable anaesthesia for the duration of the procedure
- Safe if given in overdose or by inadvertent intravenous injection

- Minimal side effects
- Easy, rapid and safe to prepare

No local anaesthetic solution fulfils the above characteristics and several may achieve reliable anaesthesia.

There is good evidence that a mix of **lidocaine 2% with adrenaline (1:200 000) and sodium bicarbonate** provides the most rapid and reliable onset of surgical anaesthesia for caesarean section.

Be cautious when checking and drawing up any drug especially in an emergency and under time pressure.

Pay particular attention to ensuring the correct dose of epidural adrenaline is provided.

Bicarbonated Lidocaine 2% + 1:200,000 adrenaline (Fizzy Lignocaine)

- Mix 17 ml 2% lidocaine + 1 ml **ADRENALINE 1:10,000 (100 mcg)** + 2 ml 8.4% sodium bicarbonate
- **NOTE: This solution requires preparation of a number of different drugs – pay special attention when preparing and ensure appropriate concentration of adrenaline is used**
- **Aspirate the epidural** catheter before administration to check for blood and CSF (catheter migration)
- Give up to 20ml in 5 ml boluses
- Check sensory level and BP as usual

Alternative top up solutions include:

Levo-bupivacaine 5 mg/ml (0.5%) or bupivacaine 0.5%

These solutions may be useful in severe pre-eclampsia when adrenaline containing solutions may potentially raise BP further.

If using these drugs give up to 20ml in 5 ml boluses.

Should the anaesthetist commence an epidural top-up in the labour room if emergency caesarean section is required?

It is acceptable to start topping up the epidural in the labour room should a woman require emergency caesarean section provided you:

- Aspirate the epidural catheter before top-up to exclude catheter migration
- Stay with the woman at all times
- Assist with the rapid transfer to theatre where the woman can be more safely observed and monitored

Addition of fentanyl (5mcg/ml) may be useful if the epidural has recently been sited. There is little evidence to suggest a benefit to adding fentanyl to a top-up mix for caesarean section if the woman has had the labour epidural in progress for some

time since a large amount of epidural fentanyl will already be present following the infusion of low-dose bupivacaine and fentanyl solution.

Administration of epidural diamorphine for post-op analgesia

At the end of the caesarean section, just prior to epidural catheter removal, give epidural diamorphine 3mg or morphine 2-3mg:

- Dilute 5 mg diamorphine in 5 ml 0.9% saline (making a solution of diamorphine 1mg/ml), discard 2 ml and giving the remaining 3 ml. Flush the epidural catheter with 5 ml 0.9% saline and then remove the epidural catheter ensuring tip is intact and document its removal on anaesthetic chart.
- Alternatively 2mg (<60kG), 3 mg (>60kG) of morphine using the PF 1mg/ml morphine

The first dose of postoperative dalteparin (Fragmin) should be prescribed for 4 hours <u>after removal</u> of the epidural catheter.

Intraoperative vasopressor use

There is good evidence that prophylactic vasopressor infusion reduces the incidence and side effects of spinal-induced hypotension.

Alpha-agonists (phenylephrine and metaraminol) are more effective in maintaining blood pressure and cause less fetal acidosis then ephedrine.

Phenylephrine given by infusion is the first line drug for preventing and treating spinal-induced hypotension.

Phenylephrine preparation and administration

- Mix 10mg phenylephrine (1 amp) in a 100 ml bag of 0.9% Saline. This makes a 100 mcg/ml dilution
- Draw up 20 ml of this solution into a 20ml syringe and place in a Graseby syringe driver
- Connect the phenylephrine infusion giving set to an anti-reflux ‘Y connector’ connected to the woman’s cannula
- Start the phenylephrine infusion at 30ml/hr (50 mcg/min) immediately following intrathecal injection
- Titrate infusion up or down by 5 ml/hr increments depending on blood pressure
- Beware of a reflex bradycardia when using phenylephrine especially in women with a low resting heart rate. Ensure that atropine, ephedrine and glycopyrrolate are to hand
- If a significant sympathetic block develops (hypotension and bradycardia, administration of ephedrine, glycopyrronium or atropine is preferable to further increasing the dose of phenylephrine

Phenylephrine may be administered as an infusion or as boluses for the management of hypotension following spinal anaesthesia. Increments of 50 - 100 mcg (0.5 – 1ml of the above dilution) have been used as bolus doses. Again, beware risk of reflex bradycardia.

Surprisingly large doses of phenylephrine (over 1mg) have been administered during caesarean section with no evidence of clinical or biochemical harm to the neonate.

Current evidence suggests aggressive treatment of hypotension aiming to maintain blood pressure close to pre-spinal levels using an alpha-agonist is best practice.

Ephedrine

Although administration of ephedrine to avoid or treat spinal-induced hypotension has been associated with lower umbilical artery pH compared with a pure alpha-receptor, it remains a useful drug in these circumstances.

A pre-filled syringe of ephedrine 3mg/ml should be close to hand when providing regional anaesthesia for caesarean section and it is especially useful in profound sympathetic block causing bradycardia and hypotension.

Vasopressor use following spinal anaesthesia in women with pre-eclampsia

Because of the nature of this condition (vasoconstricted circulation, high levels of circulating maternal catecholamines etc) some women with pre-eclampsia may be protected from spinal-induced hypotension and prophylactic administration of vasopressor in these women may lead to **hypertension** and potential harm.

The response to spinal anaesthesia in these women is not consistent so be cautious with vasopressor use and monitor blood pressure closely

Block Testing for caesarean section

All patients with a regional anaesthetic technique for caesarean section must have an adequate block check prior to commencement of surgery.

An adequate block is:

- **Bilateral loss of cold sensation to ice from sacral levels to T4, *and***
- **Bilateral loss of light touch sensation to T6 (xiphoid sternum)**

A significant motor block (absence of straight leg raise) is a reassuring feature of an adequate block.

Loss of needle-stick sensation over the incision site may provide further reassurance to the patient.

The level and description of block must be clearly documented on the anaesthetic chart.

Check the sacral dermatomes are blocked preferably by assessing the back of the thigh. The foot has crossover with lumbar dermatomes and not as reliable.

In some emergency situations, where a block is rising but has not yet achieved the above standards, commencing surgery may be appropriate provided there is good evidence a block is present and progressing appropriately. The patient should be informed of the need to expedite delivery, the possibility of discomfort and potential for GA and should give her agreement to proceeding.

Management of Intraoperative pain under neuraxial anaesthesia

Intraoperative pain under neuraxial anaesthesia may occur and can be distressing.

Pain experienced during caesarean section under spinal or epidural anaesthesia is a common cause of complaint and litigation and should be taken seriously.

In all women prior discussion and documentation of the risk of regional block failure and conversion to general anaesthesia must take place.

Should pain occur, clear contemporaneous documentation on the anaesthetic chart of its timing and its management is imperative.

Avoid topping up an inadequate labour epidural that is likely to fail in surgery.

If surgery is prolonged and a CSE is sited, top-up the epidural before pain occurs (45 minutes after CSE sited if uterus is still not closed).

Ensure adequate sensory block is present prior to commencing surgery.

If the woman experiences significant pain at the start of surgery measures to manage the pain are unlikely to work and a general anaesthetic should be offered unless the block is still rising and there is time to wait or an epidural is in situ that has not been fully topped-up.

Should pain occur, provide reassurance, stop surgery and treat the pain (see below) - conversion to a general anaesthetic may be required and should be offered.

In all cases of intraoperative pain under regional anaesthesia:

- **Clearly document on the anaesthetic chart the nature and timing of pain, what interventions were offered, administered or declined**
- **Ensure appropriate follow up and debrief takes place and referral to counselling if required**

Interventions to manage intraoperative pain include the following:

- Provide reassurance. If delivered, bringing the baby to the mother may help moderate pain
- Local anaesthetic boluses via epidural if in-situ (e.g. 5ml boluses 2% lidocaine with adrenaline 1:200 000 or 5 ml boluses levobupivacaine 0.5%)
- If the epidural has been fully topped-up administer an epidural opioid e.g. diamorphine 3mg (see above for preparation and administration)
- Administer a mix of nitrous oxide and oxygen (50:50) via the face mask of the anaesthetic machine and breathing system
- Intravenous boluses of alfentanil e.g. 100 mcg boluses
- Intravenous bolus of diamorphine – 2.5 mg
- If opioids are given prior to delivery of the fetus the neonatal doctor must be informed.

- Avoid benzodiazepines – women may find the associated amnesia distressing if unable to recall events around delivery.
- Offer a general anaesthetic.

Women who experience significant pain at caesarean section should be handed over to a consultant obstetric anaesthetist. A later debrief should be offered if necessary and in some cases referral for counselling should be considered. Consider booking an appointment in the High Risk clinic. Should the woman decline counselling, it should be documented that counselling was offered but declined.

Referrals to the Trust obstetric counselling service can be made by contacting:
The Women's Health Counsellors on: **01392 406678**

References:

Hillyard SG, Bate TE, Corcoran TB, Pach MJ, O'Sullivan G. Extending epidural analgesia for emergency caesarean section: a meta-analysis. BJA 2011; 107: 668-78

Malhotra S, Yentis SM, Lucas N. Extending epidural analgesia for emergency caesarean section. BJA 2012; 108: 879-80

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Ngan Kee WD.Comparison of phenylephrine infusion regimes for maintaining maternal blood pressure during spinal for caesarean section. BJA 2004; 92: 469-474

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E. Roofthooft,G. P. Joshi,N. Rawal,M. Van de Velde. PROSPECT guideline for elective caesarean section: updated systematic review and procedure-specific postoperative pain management recommendations Anaesthesia May 2021 Volume76, Issue5 Pages 665-68

7 Principles of Obstetric General Anaesthesia

Although the use of general anaesthesia in obstetrics has declined steadily over the last 15 years, there are important indications for its use and suitable preparation and precautions must be taken to ensure that the risks to mother and baby are minimised.

The anatomical and physiological changes of pregnancy that make obstetric general anaesthesia potentially challenging include:

- Rapid desaturation due to;
  - ↓ Maternal FRC
  - ↑ Maternal O_2 consumption

- High risk of regurgitation of gastric contents and potential aspiration:
  - Delayed gastric emptying (especially in labour and after opioid medication)
  - ↑ Maternal gastric acid reflux
  - ↑ Maternal gastric acidity

- Difficult intubation and higher risk of failed intubation (~1:250) [1]
  - Enlarged breasts may impede laryngoscopy
  - Poor positioning and preparation due to emergency nature of caesarean under GA
  - Impaired performance due to stress and anxiety
  - Increased risk in obese

- Aortocaval compression

- Higher incidence of awareness
  - Rapid redistribution of IV induction agent
  - Rapid commencement of surgery before inhalational agent may have reached effect site
  - Reduced dosing of volatile agent due to concern of impairing uterine tone

Always call for help early if concerned that general anaesthesia may be problematic

<u>Obstetric General Anaesthesia: A recipe</u>

Note – modifications to the approach below may be required in certain situations (e.g. pre-eclampsia)

Please refer to the GA RSI checklist (see appendix)

Pre-assess

- Take a relevant history and make an assessment of the airway

Acid prophylaxis

- Administer antacid prophylaxis (Omeprazole 20 mg orally, Sodium Citrate 30ml)

- Give 30 ml Sodium Citrate just prior to pre-oxygenation

Position

- Optimal positioning to minimise problems at laryngoscopy
- Consider ramped position if obese (see Figure 1, page 23)
- Provide left lateral tilt until after delivery

Plan

- Ensure alternative airway equipment is available and to hand (e.g. second generation LMA)
- Ensure the CMAC laryngoscope is in theatre and to hand. Strongly consider this as first line.
- Ask yourself the questions:
- What is my plan if I fail to intubate?
- What size and type of LMA will I insert? Typically 2nd generation LMA.
- Where is the LMA that I might need?
- What is my plan if I fail to intubate but can ventilate – wake up or continue? (see failed intubation algorithm)

Pre-oxygenate

- 3 min tidal volume breathing or 12 deep breaths with tight fitting face mask with fresh gas flow rates >10L/min (ensures faster pre-oxygenation due to reduced entrainment of room air)
- Head-up tilt increases FRC and may prolong the time to desaturation
- Look at EtO2 during pre-oxygenation – aim for >0.85
- **During pre-oxygenation run through obstetric GA check list – see below**

Induction

- Rapid sequence induction using propofol (2.5mg/kg)
- If significant hypovolaemia and/or severe sepsis use ketamine (1.5mg/kg)
- Give suxamethonium (unless contraindicated) (1.5mg/kg)
- If using rocuronium give 1mg/kg
    - Have sugamadex to hand and be aware of dose
    - **2-4mg/kg – for routine reversal**
    - **16mg/kg for immediate reversal** of rocuronium-induced blockade
- Alfentanil 10 - 20 mcg/kg can be administered at the time of intubation and has been shown not to have clinically significant effects on neonatal APGAR scores.

Maintenance

- Administer 50% oxygen in nitrous oxide – nitrous oxide is advisable in obstetric GA to reduce risk of awareness
- Use either sevoflurane or isoflurane (sevoflurane may have advantages in terms of faster onset (potential reduced risk of awareness) and more rapid offset
- Ensure adequate anaesthesia prior to skin incision
- Aim for MAC >0.8

- A single dose of suxamethonium will often provide adequate muscle relaxation for the entire case. If further muscle relaxation is required give a small dose of non-depolarising relaxant (e.g. atracurium 10mg

Recovery

- Following delivery give an opioid e.g. morphine - total dose 10-20mg
- **Check neuromuscular blockade and reverse if necessary**

  Note: Magnesium may potentiate neuromuscular block – be cautious in women administered magnesium (e.g. pre-eclampsia)
- Be aware that prolonged block may occur after a single dose suxamethonium
- If high risk of regurgitation (e.g. obese, recent meal, caesarean in labour) consider emptying the stomach prior to extubation by passing a wide bore oro-gastric tube (do not pass nasally due to risk of bleeding from engorged nasal vessels) – This is a recommendation from the Confidential Enquiries into Maternal Deaths
- If able and time permits administer TAP blocks under US guidance
  - Ensure sterility
  - Give 20 ml 0.25% levobupivacaine to both sides
- Awake extubation in the lateral position. **Beware – there have been recent cases of regurgitation and aspiration following extubation and recovery from general anaesthesia**
- Ensure appropriate staff are present to recover the patient. If not, you must stay with the woman until conscious and stable.

Care of women with hypertensive disease who require general anaesthesia

- In women with raised blood pressure (e.g. pre-eclampsia) and those with cerebrovascular disease – it is essential to obtund the pressor response to laryngoscopy and intubation. Suggested drugs include:
  - Alfentanil 10 - 20 mcg/kg (this is probably the easiest to administer)
  - Magnesium – 40mg/kg
  - Remifentanil – 1 mcg/kg
- If administering an opioid at induction, inform the paediatrician of possible neonatal respiratory depression.

8 Failed Obstetric Intubation

Failed intubation occurs more commonly in the obstetric patient compared to the non-obstetric patient for a host of anatomical, physiological and situational reasons.

The rate of failed obstetric intubation is around 1:250.

The risk of failed intubation is increased with obesity, increased maternal age and increasing Mallampati score.

The higher rate in obstetrics is undoubtedly influenced by the emergency, high-pressure situation in which most obstetric general anaesthetics takes place.

Examine, assess and document the airway in all cases presenting to theatre for either regional or general anaesthesia.

Identify those at risk for failed intubation and summon senior help.

Remember that regional blocks may fail and general anaesthesia may be required midway through a case. Be prepared.

Elevate (Ramp) the upper back of obese patients using pillows or by ramping the operating table.

Figure 1

Obstetric Anaesthetists Association and Difficult Airway Society issued guidelines for failed obstetric intubation in 2015. Appendix 1.

We do have an Oxford Pillow that may be used for optimising position for intubation. It resides in CWH1 and should be considered for a potentially difficult GA LSCS. Bear in mind the pelvic flexion can make it more difficult for the obstetricians to deliver the baby. It may need to be removed, if possible, before knife to skin.

In all cases consider contingency plans in event of failed intubation prior to induction

Women do not die from failure to intubate but from failure to oxygenate

- Prolonged attempts at intubation are likely to lead to maternal hypoxia and airway trauma and should be avoided
- Oxygenation and ventilation by alternative airway techniques is key

9 Pulmonary Aspiration

A host of measures in obstetric anaesthesia are used to prevent pulmonary aspiration of gastric contents and limit the damage should it occur.

Regurgitation and pulmonary aspiration may occur at induction of general anaesthesia, at extubation and recovery from general anaesthesia and in cases of maternal collapse.

If aspiration is recognised at the time, the lungs should be sucked out immediately, prior to ventilation if possible. There is no evidence that any specific measure influences the subsequent development of problems and the main aspect of management is to maintain oxygenation and treat any other complications that result.

If no signs of respiratory problems develop, such as pulmonary oedema and oxygenation is not impaired, the patient can be extubated and monitored closely. If there is respiratory compromise do not extubate but continue ventilation with PEEP and contact the ICU for advice and to arrange transfer.

Measures to limit the risk of aspiration have been detailed earlier and include:

- Minimising oral intake in high risk labour
- Identification of those at high risk of intervention and prophylactic prescription of oral omeprazole 20mg
- Administration of prophylactic omeprazole to all women requiring an intervention under anaesthesia (GA or RA)
- Use of measures to limit the risk of regurgitation/aspiration – RSI and head-up-tilt where appropriate and recovery with patient in the lateral position
- Aspiration of stomach contents of women at high risk of regurgitation before extubation following caesarean delivery under GA
- Women at higher risk of aspiration include the following:
    - The morbidly obese, earlier administration of opioids, those in labour, those known to have a full stomach

References:

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Post-Partum Analgesia

The obstetric anaesthetist has a responsibility for prescribing and managing post-partum analgesia for women who have an operative intervention.

Regular Analgesia

- Prescribe Paracetamol 1g oral QDS regularly to all
- Prescribe a NSAID to all unless contraindicated, Ibuprofen 400 mg QDS
- Omit ibuprofen for 18 hours if ibuprofen has been given.
- Dihydrocodeine 30mg QDS
- **NSAIDs are contraindicated in pre-eclampsia, major/on-going haemorrhage, renal disease and sepsis**

As Required Analgesia

- Prescribe Oramorph 10 – 30 mg 3 hourly

Other as required analgesia

- Tramadol 50-100mg oral PRN to a total of 400mg per day
  The manufacturer advises avoid when breast-feeding although amount probably too small to be harmful. **Reserve for occasional cases or if the baby is bottle-fed**

As Required Antiemetics

- Ondansetron 4 mg IV 8 hourly
- Cyclizine 50 mg IM/IV/PO 8 hourly

Drugs for pruritus

- First line: Naloxone 40-80 mcg iv (dilute 400mcg in 10 mls and give 1 or 2 mls).
- Second line: Chlorpheniramine 4mg QDS but care in breast feeding.

Others

- Movicol BD to counteract the side effects of dihydrocodeine

Post-partum Patient-Controlled Analgesia (PCA)

Women with a PCA cannot be managed on the post-natal ward and must remain on labour ward.

Women who have a caesarean section under general anaesthesia do not routinely require a PCA.

A PCA can often be avoided if appropriate multimodal analgesia is provided and appropriate doses of intravenous morphine administered at or immediately following caesarean section under general anaesthesia.

If despite this the woman has significant pain, a PCA should be provided and the woman managed on the labour ward.

Prescription for Morphine PCA

- 50mg morphine in 50ml saline (prefilled syringes prepared by pharmacy)
- 1mg IV bolus
- 5 minute lockout
- No background infusion

Prescription for Fentanyl PCA

- Fentanyl 500 mcg added to 40 ml 0.9% NaCl to make a 50 ml solution containing 10mcg/ml fentanyl
- Prescribe a 10mcg (1ml) bolus with 5 minute lock out and no background infusion

TAP Blocks

TAP blocks have been shown to reduce morphine consumption and improve postoperative pain following caesarean section under GA.

They may also be useful if a caesarean section was performed under spinal without a long-acting opiate:

- Use only if competent.
- Perform after delivery.
- Use full aseptic precautions. Clean skin with Chloroprep sticks.
- Use a 100mm nerve stimulating needle under US guidance and place 20 ml of 0.25% bupivacaine on both sides.

References:

MHRA drug safety update (codeine) vol 6, issue 12, July 2013: A1

Tan TT, Teoh WH, Woo DC, Ocampo CE, Shah MK, Sia AT. A randomised trial of analgesic efficacy of ultrasound-guided transversus abdominis plane block after caesarean delivery under general anaesthesia. Eur J Anaesthesiol 2012; 29: 88-94

10 Thrombo-Prophylaxis

VTE risk and possible need for post-operative thrombo-prophylaxis should be discussed at 'Time Out' prior to all obstetric surgical interventions

Post-Caesarean Section thrombo-prophylaxis

Low and High Risk Women

- Wherever possible, women should have graduated elastic compression stockings (TEDS) applied pre-operatively. If the woman is not wearing TEDs, this should not delay transfer for a category 1 caesarean in which case they can be applied once in theatre or after surgery.
- Some women may require calf compression devices during and after caesarean section (see below)
- Unless contra-indicated, all women undergoing caesarean section should receive prophylactic dalteparin (fragmin).
- The dose of LMWH (dalteparin) depends on the booking weight:

Weight (Kg)	Standard Dalteparin Dose (International Units)
<50	2500 units daily, S/C
50-90	5000 units daily, S/C
91-130	7500 units daily S/C
131-170	10,000 units daily S/C (given in 2 divided doses)
>170	75 units/kg/day (given in 2 divided doses)

- The first dose should be given 4 hours after any insertion of an epidural or spinal, or 4 hours after removal of an epidural catheter.
- Post-operatively the LMWH should continue for at least 10 days.
- If prescribed once daily the LMWH should be given at 1400.
- If prescribed twice daily the LMWH should be given at 0800 and 2000.
- To enable TTO prescription, ensure the correct duration of treatment is ordered on MyCare. Typically 10 days post LSCS.
- Epidural catheters should be removed prior to transfer to the postnatal ward. In exceptional circumstances an epidural catheter may remain in-situ post-delivery, e.g. following complex surgery in which postoperative epidural analgesia is indicated. In these cases, women must remain on the labour ward while epidural analgesia is provided.
- Any woman who is an in-patient prior to her caesarean section should, unless contra-indicated, receive prophylactic dalteparin (Fragmin) 5,000 IU no later than 1800 the day before surgery.

- For patients coming to theatre but not undergoing LSCS please refer to obstetric VTE guideline for individual risk assessment.

Management of women already on LMWH

- Women already receiving antenatal LMWH should be advised to withhold the next dose when they go into labour

Women on prophylactic dose LMWH (e.g. dalteparin 5,000 IU once daily):

- Wait 12 hours after last dose before inserting a spinal / epidural catheter
- Allow 4 hours after insertion / removal of catheter before next dose

Women on treatment dose LMWH

- Wait 24 hours after last dose of LMWH before spinal / epidural catheter insertion or removal
- Allow 4 hours after insertion / removal before next dose
- If bloody tap on epidural insertion consider omitting the next dose

Women on prophylactic dose LMWH <u>and</u> aspirin 75mg:

- These women are at increased risk 12-24hr
- Treat as for therapeutic dose

Women on intravenous un-fractionated heparin infusion

- Managing these women can be challenging. Liaise closely with haematology and follow guidance on IAN
- The infusion should be stopped for 4 hours or until the APTT ratio has returned to normal prior to epidural / spinal insertion or removal. TEG is useful in identifying heparin effect – if the difference between R(CK) and R(CKH) is <2 mins this indicates no effect of heparin on coagulation
- Wait 4 hour after insertion / removal before restarting infusion

Individual cases that require anticoagulation (e.g. women with a prosthetic heart valve) should be discussed with a senior anaesthetist, obstetrician and haematologist.

- After delivery be vigilant for signs of vertebral canal haematoma
- Women should be advised to seek medical help if symptoms of weakness, numbness, acute back pain, or bowel / bladder dysfunction develops over the next 48 hours
- Suspected vertebral canal haematoma must be promptly investigated by neuraxial imaging

References:

Regional anaesthesia and patients with abnormalities of coagulation. AAGBI Guideline. November 2013

Reducing the Risk of Venous Thromboembolism during Pregnancy and the Puerperium, Green Top Guideline No. 37a, RCOG April 2015

11 Calf Compression Device (e.g. Flowtron®) use during and after Caesarean Section

The following patients fall into a higher risk group. The use of perioperative calf compression devices should be considered in women with the following risk factors:

Medical risk factors

- Obesity (booking BMI > 35)
- Diabetes
- **Medication causing increased risk e.g. long-term high dose steroids, cytotoxic drugs**

Neurological risk factors

- Paraplegia / hemiplegia
- Chronic neurological disorder e.g. multiple sclerosis with limited mobility
- Other cases with significantly impaired mobility

Obstetric risk factors

- Multiple pregnancies
- In-patients with reduced mobility
- Women with severe preeclampsia and other critically unwell patients likely to be immobile for some time

12 Emergency Second Obstetric Theatre

In the event of needing a second obstetric theatre for an emergency (Cat 1 LSCS/major PPH) the first option is to use either theatre 2 or 3. If unable then the theatre 2 anaesthetic room will be used.

It is the Labour ward Co-ordinators responsibility to organise the second theatre team.

- Surgeon – Contact the On call Consultant immediately – if already present liaise with them who next to call
- Scrub team - Liaise with scrub team in theatre 1 +/- Bleep 801 (main theatre team) – they will also assist with organising additional ODP
- Anaesthetist – Obstetric on call Anaesthetist will be aware of other staff available in unit. On call Anaesthetist Consultant can also be contacted via switchboard

It is the responsibility of all staff members who may be involved with a category one LSCS, in this unusual location to familiarise themselves with the points below and to look at the small space available.

As the room will only be used in life threatening emergencies, it is likely the patient will have a general anaesthetic and hence a CTG machine will not be required and the patient's partner can wait in the delivery room.

Additional process of Cat 1 LSCS in anaesthetic room one:

Anaesthetist and ODP

- Contact the co-ordinating ODP on bleep 415 to inform them of what is going on so they can either come directly or allocate extra staff to assist
- Patient must be transferred to a trolley to allow for more space in the anaesthetic room. Remember to insert wedge under right hip. Assist with transferring patient to theatre whilst taking history.
- Following induction of anaesthesia – move operating trolley to the right to give more room for theatre scrub nurse
- Space is limited and you will need to move to patients left side to allow midwife through door who will then take the baby to the neonatal team in the corridor outside the anaesthetic room.

Midwife responsibility

- 2222 and state cat 1 caesarean section anaesthetic room 2, labour ward
- Collect a day case operating trolley (with canvass, wedge and pat slide) to the patient in the delivery room. Transfer the patient onto the theatre operating trolley in the **delivery room** and then take into anaesthetic room. You cannot fit a delivery bed into this room.
- Request that matron/another midwife moves the resuscitaire from the delivery room to outside the anaesthetic room and plugs it in and turns on (see picture 1)
- Obtain portable light (delivery rooms) and transfer to anaesthetic room

- After handover of care to theatre staff, ensure resuscitaire is plugged in and turned on (see picture 1), then put on a sterile gown and gloves in order to take baby from surgeon to resuscitaire. **You do not need to formally scrub up.**

Theatre scrub nurse

- Obtain spare suction machine and Olympus Diathermy machine from theatre trolley bay
- Scrub up in theatre 2 scrub room
- Open pack in the anaesthetic room and proceed as normal
- Once surgical assistance in place ensure diathermy and suction plugged in safely

Surgeon

- Scrub up in theatre 2
- Wear 2 pairs of sterile gloves so indwelling catheter can be inserted – this cannot be done by midwife due to space issues
- Once baby is delivered turn to **left** and hand to midwife who should be wearing a sterile gown and gloves.

Surgical Assistant

- Scrub up in theatre 2
- Move to position on left side of patient as quickly as possible so theatre scrub nurse can move her equipment closer.

Neonatal Team

- Ensure you are in the right place
- Once the patient has been moved into the anaesthetic room, move resuscitaire over to plug socket (pic 1), plug in and turn on. This may not have been done already by the midwife
- The midwife will carry the baby to resuscitaire
- In the event of prolonged resuscitation, be aware that you will be using cylinder oxygen and hence moving to the neonatal unit may be advisable.

Opening an Obstetric Theatre in Main Theatres

In the event of needing to move the obstetric theatre, for either an elective or emergency LSCS, the following sequence should run in order of priority/suitability:

1. Theatre 2 or 3 in CWH.
2. Theatre 2 anaesthetic room in an emergency.
3. If CWH is not available at all, then main theatres should be used (ideally Theatre 5).

It is the responsibility of all staff members who may be involved with the LSCS, in this different location, to familiarise themselves with the **Clinical Guideline for Caesarean Section In General Theatres.**

13 Anaesthesia for Operative Vaginal Delivery

Regional anaesthesia is preferable for operative vaginal delivery since women are still able to push even with a sensory and motor block, and this may contribute to a successful vaginal delivery.

Use the same consent, precautions and techniques as for regional anaesthesia for caesarean section including.

If a block fails, is contraindicated or is declined, the obstetricians may be able to perform the instrumental delivery under a pudendal block. Failing this a general anaesthetic may be required though this is not ideal for operative vaginal delivery.

If a woman is transferred to theatre for a trial of operative delivery, there is a chance caesarean delivery will be required in the event of failure. Therefore, when providing a block for operative vaginal delivery, administer an epidural or spinal anaesthetic that is adequate for caesarean section. Be aware that if instrumental delivery fails the obstetrician may rapidly proceed to caesarean section.

The trial of instrumental delivery need not be delayed for the block to reach a level suitable for a caesarean section but can commence once the woman is comfortable and the block is clearly effective and rising.

The same preoperative checks, antacid prophylaxis and IV access should be established as for a caesarean section. Even though the procedure is done in the lithotomy position, left lateral tilt should be used until delivery.

Caesarean section following a failed instrumental delivery is classified as Category 1.

- **Ensure a block suitable for a caesarean section is achieved when providing regional anaesthesia for instrumental delivery in theatre**

14 Anaesthesia for Other Operative Procedures

Aside from caesarean section and operative vaginal delivery anaesthesia may be required for other operative procedures including:

- Insertion of a cervical suture
- Manual removal of retained placenta
- Suturing of perineal tears
- EUA and evacuation of retained uterine products
- Laparotomy

All women from 16 weeks of pregnancy to 2 days postpartum should be considered to have delayed gastric emptying. Women should have H_2 antagonists (if time permits) and sodium citrate. If general anaesthesia is chosen, a rapid sequence induction should be performed.

Regional or general anaesthesia can be used for the above procedures and the choice of anaesthetic technique will be influenced by individual factors.

Post-partum procedures may be associated with substantial haemorrhage and hypovolaemia and this should be anticipated.

In the presence of substantial and/or continual haemorrhage, regional anaesthesia is usually avoided because of the resultant sympathetic block.

Aortocaval compression by the gravid uterus is no longer an issue in post-partum procedures therefore lateral tilt is not required.

General anaesthesia

Women are still at risk of acid aspiration for at least 48 hours post-partum and the standard precautions should be taken (ranitidine, sodium citrate, RSI, lateral recovery).

Regional anaesthesia

The physiological and anatomical changes in the spinal canal associated with pregnancy are abolished almost immediately after delivery, and doses of local anaesthetic required for epidural or spinal block rapidly revert to non-pregnant (higher) values.

The sensory block required for the above procedures, however is significantly lower than that required for a caesarean section (excluding laparotomy) so relatively low doses of local anaesthetic are still effective.

2% Prilocaine is routinely used for manual removal of placenta and simple perineal tears. The duration of anaesthesia is 75-90 minutes so worth checking with the obstetrician if they think it is complicated or may take longer. These patients just need in-out urinary catheters unless there is an obstetric indication for the catheter.

See appendix for prilocaine poster and dosage.

If in doubt about the length of procedure then use 2 mls heavy bupivacaine 0.5%.

Manual removal of placenta

A single shot spinal is often appropriate unless an effective labour epidural is in-situ in which case it can be topped up.

A block to T 10 is recommended and can be provided by incremental top-up of the epidural (using drugs discussed above for caesarean), prilocaine or hyperbaric bupivacaine 0.5% (2mls). Additional spinal opioid is not usually required.

Retained placenta is associated with post-partum haemorrhage

- **Be alert for signs of bleeding and inadequate uterine contraction after placental delivery**
- **Ensure appropriate iv access and blood has been grouped and saved**
- **Uterotonics including a syntocinon infusion are often required**

Insertion of Cervical Suture

This is often performed at the end of the first trimester/early second trimester.

Regional anaesthesia - single shot spinal either 2 mls 2% heavy prilocaine or hyperbaric bupivacaine 0.5% heavy with no additional opioid. See appendix for prilocaine use poster.

General anaesthesia can also be used. Although recent concerns with in-utero fetal exposure to general anaesthesia have recently been highlighted and regional anaesthesia may be advisable.

Some obstetricians may prefer general anaesthesia since uterine relaxation may aid the procedure. In particular if there is predicted to be difficult access to the cervix and steep head down is required. It is useful to discuss the anaesthetic plain in advance.

EUA and evacuation of uterus

This is often performed because of ongoing post-delivery bleeding.

Regional anaesthesia is appropriate as long as the woman is not hypovolaemic. Use 2% heavy prilocaine (if thought uncomplicated) or heavy bupivacaine 0.5%. See appendix for dosage.

Alternatively use a general anaesthetic with rapid sequence induction if less than 48 hours post-delivery.

Laparotomy

Laparotomy may be required for ongoing post-partum haemorrhage and general anaesthesia will commonly be required.

Ensure appropriate large bore iv access, consider invasive arterial monitoring, intraoperative cell salvage and call for senior help.

15 Epidural Analgesia for Labour Pain

Epidural analgesia is the most effective form of labour pain relief.

It may be particularly useful in women whose labour is likely to be long or difficult, when operative delivery is likely and in women in which general anaesthesia may be challenging, e.g. morbid obesity, likely difficult intubation, and cardiac disease.

In the first stage of labour, pain from uterine contractions and cervical dilatation is conducted through spinal nerve roots T10-T12 via the paracervical plexus. For the second stage of labour, innervation of the pelvis and perineum is via the pudendal nerve to spinal roots S2-S4.

Properties of the ideal labour epidural include:

- Provision of sensory blockade adequate for both first and second stages of labour
- Avoidance of motor block to enable mobility and pushing and avoid risk of pressure sores
- The ability to top up and convert the labour epidural to an operative epidural
- A low risk of procedure-related complications

In practice, the L2/3 or L3/4 interspace is normally used for epidural catheter insertion and the necessary block achieved by the spread of an appropriate dose and volume of drugs.

Addition of opioid to the epidural mixture allows the local anaesthetic concentration to be reduced while maintaining good quality analgesia, thus reducing unwanted motor blockade.

Consent

However distressed the mother may be in labour, consent must be obtained before undertaking epidural analgesia.

Consent for an epidural **must** be documented on the epidural form.

Women should be given the OAA epidural information card to read prior to discussion with the anaesthetist. This information is available in a number of different languages.

Complication risks and their frequency of occurrence should be discussed in line with the information from the OAA (see Appendix 1)

The extent to which all potential complications are discussed will depend on individual circumstances, but **the following risks must be explained**:

- Incomplete analgesia (incidence up to 1:5) and possible need to re-site epidural (about 5%)
- Motor block
- Drop in blood pressure
- Headache due to accidental dural puncture (incidence 1 in 100)
- Nerve injury both temporary (~1:1,000) and permanent (~1:13,000)
- Risk of infection (1:50,000) or haematoma (1:170,000)

You should consider discussing the following if appropriate:

- A possible association with a higher rate of operative delivery. This remains controversial and is less evident with the use of low-dose epidural infusions.
- Urinary retention
- Good evidence that there is no association with long-term backache

Preparation

The following measures must be taken before epidural insertion:

- OAA Epidural information card seen by patient. **Appendix 7.**
- Maternal consent
- Measure blood pressure
- Establish large bore IV access (16G)
- Position patient in either the lateral or sitting position
- Full aseptic precautions – gown, gloves, hat and face-mask
- Preparation of the area with 2% Chlorhexidine and alcohol (Chloraprep®), applied and allowed to dry
- Infiltration of the skin and superficial layers with local anaesthetic.

Chlorhexidine may cause serious neurological damage if in contact with neuraxial structures

- **Apply to the skin *prior* to preparation of epidural/spinal equipment**
- **Avoid contamination of gloves and change if necessary**

Labour Epidural Technique

- Loss of resistance to saline should be used
- 3 – 4 cm of catheter should be left in the epidural space. Leave 1-2 cm more if obese. Be mindful that sometimes, especially in the obese, when the woman extends the back after being positioned for epidural insertion, the catheter may migrate in. Should this occur, secure at this length rather than withdrawing the catheter and document that this has been done. This may prevent migration of the catheter out of the space when the woman flexes the back again.
- Secure with a Lockit™ and adequate sterile dressings
- Aspirate catheter for blood and CSF prior to first dose. Attach the giving set to the 250ml 0.1% bupivacaine + 2mcg/mL fentanyl bag, prime the set (see below) and connect to the epidural filter. This is the responsibility of the attending anaesthetist.
- A test dose of 8 mg bupivacaine should be given – **Give this using the low dose mix i.e. 8 ml of bupivacaine 0.1% +2mcg/ml fentanyl through the epidural pump**
- Following the test dose, check maternal blood pressure and confirm the absence of motor block and rapid resolution of pain due to intrathecal administration
- Give a further 7ml of low dose mix (15 ml in total)

- The maternal pulse rate (PR), blood pressure (BP) and fetal heart rate (FHR) are recorded every 5 mins for 25 mins and then recorded on the epidural observation chart
- Remember to sign for the epidural low-dose mix in the controlled drug book that the midwife should bring to the labour room
- Return in 10 – 15 minutes to assess the quality of the epidural block

There is evidence that reducing pain whilst maintaining motor function improves both maternal satisfaction and the outcome of labour. By avoiding high concentrations of

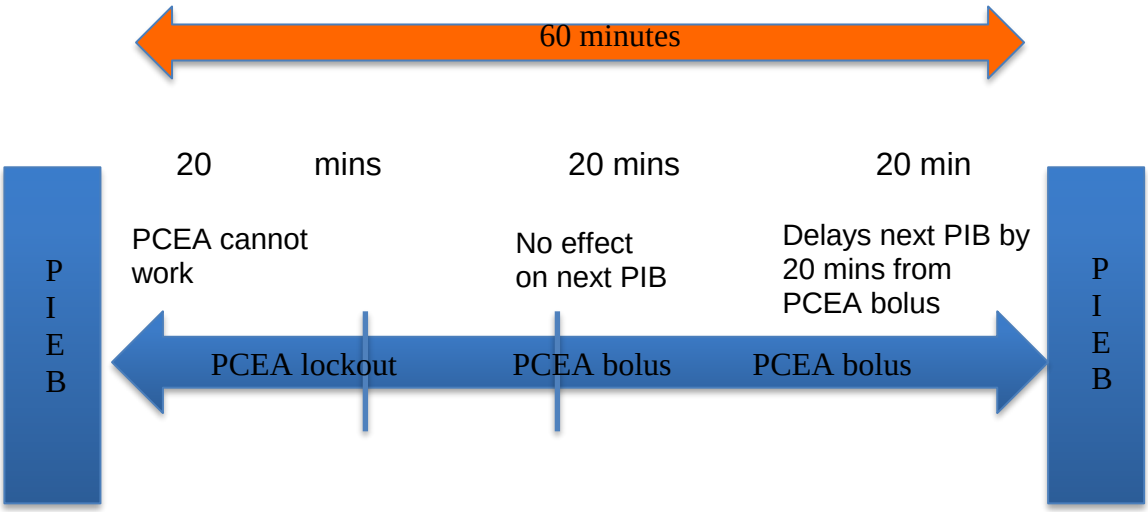
local anaesthetic, motor block is minimised, cardiovascular instability reduced and the ability to push in the second stage maintained.

For epidural analgesia in labour, minimise motor block by using the low-dose mix of 0.1% bupivacaine with 2 mcg/ml fentanyl wherever possible

PIEB Epidural Pump setup

PIEB mandatory bolus - 7 mls 0.1% Levobupivacaine and 2mcg/ml Fentanyl
PIEB bolus interval - 60 minutes
PCEA bolus - 6 mls 0.1% Levobupivacaine and 2mcg/ml Fentanyl
PCEA lockout - 20 minutes

If PCEA bolus within 20 mins of next PIEB dose then the next PIEB is delayed for 20 mins.



The anaesthetist is responsible for priming and setting up the epidural pump and selecting the chosen protocol.

Please see the laminated sheet attached to the epidural trolley for setting up the pump.

Press start on the pump to commence the protocol. The first intermittent bolus (PIEB) will not be given for an hour. Give the woman the PCEA cord **(it will not work until 20 minutes after the pump has been started)** and explain that if the handset light is on, then the PCEA is available.

Give an explanation of how the pumps work to the woman and her birth partner is essential to ensure that the epidural has the best chance of success. This might be best done after the initial top-up has had a chance to work.

The midwife should not disconnect and give an escape dose using this protocol.

Epidural top-ups this can be done by stopping the pump and selecting a Clinician Bolus of the mixture. This is typically 5 mls followed by a further 5 mls if required. A clinician level 2 code this required for this. 0.25% bupivacaine can be given but this requires disconnection of the pump and therefore should be avoided if possible. Boluses should be given only by the attending anaesthetist and not the midwife.

Removal of the epidural catheter after delivery

Epidural analgesia should be continued until after completion of the 3rd stage and any necessary perineal repair. The epidural catheter can be removed once the 3rd stage has been completed, cardiovascular stability confirmed, and there is no evidence of any bleeding. Occasionally the anaesthetist may ask that the epidural catheter is removed later either at a specific time or when requested.

Timing of removal is especially important in women who have been receiving or will be requiring low weight molecular heparin for thromboprophylaxis. The anaesthetist should give clear written and verbal instructions.

The tip of the epidural should be inspected for the ‘blue tip’ to show it is complete. This should be documented, together with the time of removal in the intrapartum record. If removed in the operating theatre after a procedure this should also be recorded.

The mobility assessment should be done by the midwife caring for the woman and recorded in the intrapartum document. If the woman has passed she should then be encouraged to mobilise.

Low-dose CSE technique for labour

- A low-dose CSE may be appropriate in some circumstances to establish rapid onset of pain relief in women who are nearing the end of the first stage of labour or who are unable to stay still to allow an epidural to be safely inserted.
- A low dose CSE can be performed in a delivery room.
- A low dose CSE can be provided by placing a single shot low dose spinal (25G) followed by an epidural once the woman is comfortable.

Alternatively, a needle-through-needle technique can be used (Portex-Smiths CSEcure kit)

- The low-dose spinal mix should be made up in a 2ml syringe:
  - **1ml 0.25% plain bupivacaine + 15mcg fentanyl (0.3ml) made up to 2 ml in total with 0.9% NaCl (0.7ml). This makes a 0.125% bupivacaine +15mcg fentanyl solution**
  - **Alternatively use 2 ml of the low-dose epidural mix (Bupivacaine 0.1% and fentanyl 2mcg/ml)**
- Note with the above CSE techniques the epidural catheter is not tested since a spinal block is provided first. The anaesthetist must test the catheter and give the first epidural dose e.g. 8ml low-dose mix (0.1% bupivacaine +2mcg/ml fentanyl).

Whenever injecting drugs intrathecally be cautious both in sterility and in preparation of correct dose.

Low-dose spinal anaesthesia as described above is unlikely to cause a significant fall in BP or a motor block.

Spinal doses higher than those detailed above must **not** be administered outside the operating theatre.

Experience from centres that perform high numbers of low-dose labour CSEs suggest that complications are no higher than when labour epidurals are routinely performed – including the risk of PDPH or intrathecal migration of the epidural catheter.

Troubleshooting Inadequate Epidural Analgesia

Although regional analgesia is the most effective form of pain relief in labour it is not always perfect. If asked to review an inadequate epidural block, assess the distribution of the block with ice and observe the woman during a contraction to try to establish a possible cause and solution. Consider also the parity of the woman, stage of labour and likelihood of subsequent intervention.

Improvements in pain relief can sometimes be achieved with simple manoeuvres including changing the position of the woman and withdrawing the epidural catheter rather than immediately administering a higher dose of local anaesthetic.

Determine the following:

Where is the pain or discomfort?

What is the height of the block? (Check this yourself!)

Is it equal on the left or right?

Are there any missed segments?

Is there good sacral coverage?

How far has the mother progressed through labour?

What is the orientation of the fetal head?

Is the pain related to a full bladder?

- **No effect of epidural dose on labour pain –** if after the first dose of low-dose mix there is no effect on pain and no demonstrable sensory block after 20 minutes, the catheter is unlikely to be correctly sited in the epidural space. Discuss with the woman and resite the epidural if still indicated.
- **Missed segment** – lie the woman affected side down and give a bolus of 10 ml low-dose mix (e.g. 10 ml 0.1% bupivacaine + 2mcg/ml fentanyl) or 5 +5 ml 0.25% bupivacaine. If no improvement manage as for unilateral block.
- **Unilateral block** – withdraw the catheter by 1 – 2 cm and give a further dose of low-dose mix (e.g. 10 ml 0.1% bupivacaine + 2mcg/ml fentanyl) with the woman lying with the affected side down. Leave at least 3 cm in the epidural space. If this fails to improve pain relief consider re-siting the epidural.
- **Patchy block –** consider a top-up with fentanyl 50-100 mcg in 10 ml 0.9% sodium chloride to improve spread. A stronger dose of local anaesthetic (see above) may also be required. Consider the possibility of a subdural block (see below).
- **Persistent perineal pain –** This is more common with an OP position in the later stages of labour. Give a bolus of fentanyl 50-100 mcg in 10 ml 0.9% sodium chloride or a bolus of 10 ml 0.25% bupivacaine with the woman sitting.
- **Pain breaking through a good epidural block** – consider the possibility of uterine scar rupture in women labouring with a previous caesarean section scar. Assess vital signs and request obstetric review.

Important Points

- If the above troubleshooting approaches fail to establish adequate analgesia, the epidural should be resited.
- Poor regional analgesia in labour predicts poor surgical anaesthesia should the epidural be topped up. Have a low threshold for resiting a poorly performing epidural, for example: in women at risk of caesarean section, in women where caesarean under general anaesthesia may be problematic (e.g. obesity, difficult intubation) or women where optimal pain relief in labour is necessary e.g. cardiac disease.
- Women with an epidural are at higher risk of operative delivery and should be reviewed when the shift changes so any problems with the block can be identified early and potentially rectified.

Subdural Block

An epidural catheter may rarely cause separation of the arachnoid mater from the dura and enter the subdural space. Characteristics of a subdural block include:

- An unexpectedly high level of sensory block, sometimes as high as the cervical dermatomes, occurring over 20-30 min after a top-up
- Sensory block may be patchy, often with missed segments and persisting pain
- Nasal stuffiness and Horner’s syndrome can develop
- Motor block is often minimal

Since the arachnoid is thin and easily torn, a subdural catheter may rupture through this layer following a bolus dose and a subdural block may become a spinal block, which may result in a high regional block.

Have a high index of suspicion for a block with a 'bizarre' distribution. Seek advice and resite the epidural at a different level.

If the catheter appears to be subarachnoid this should be clearly marked and explained to the patient. It can be used throughout labour but the anaesthetist must give all top-up doses. See guidance under management of dural tap.

References:

Bogod D. A sting in the tail: antiseptics and the neuraxis revisited. Anaesthesia 2012; 67: 1305-9

Holdcroft et al. Neurological complications associated with pregnancy. BJA 1995; 7: 522-526

Zwissler B. Regional anaesthesia and analgesia for labor and delivery. N Engl J Med 2003;348:1818-20

16 PCA for labour pain

Women with contraindications to regional analgesia who are unable to manage their pain using other non-epidural techniques and drugs available on the labour ward may receive an opioid PCA.

A remifentanil PCA should not be administered because there is a potential risk of maternal apnoea and continuous presence of a midwife in the room cannot be guaranteed

Fentanyl PCA

Fentanyl 500mcg (10ml) + 40 ml Normal Saline to make a 50 ml solution: 10mcg/ml

Prescribe 1 ml (10mcg) with a 5 min lock out and no background infusion

- Be aware that PCA labour analgesia is often not satisfactory and there is a risk of maternal sedation
- Ensure continuous O^2 saturations monitoring and hourly observations of heart rate, BP and sedation in line with the trust guidelines for PCA use.

PCAs may be provided in the event of a late miscarriage to facilitate delivery. Morphine PCA is suitable. In this instance an epidural may also be considered and depends on the preference of the patient following a discussion of both options.

17. Unintentional Dural Puncture

Unintentional dural puncture may occur due to the Tuohy needle or the epidural catheter.

If a dural puncture has clearly occurred as a result of the Tuohy needle (i.e. flow of clear fluid (CSF) from the Tuohy needle) there are two management options – to insert an intrathecal catheter and provide subarachnoid analgesia or remove the Tuohy needle and resite the epidural at a different level. The former option may have advantages.

Insertion of an intrathecal catheter

- Immediately replace the stylet into the Tuohy needle to minimise loss of CSF and prepare to insert the epidural catheter
- Thread the epidural catheter into the intrathecal space leaving the same length of catheter you would for epidural analgesia (~ 4cm)
- Secure the epidural catheter in the usual way
- **CLEARLY LABEL THE CATHETER AND FILTER AS INTRATHECAL AND COMMUNICATE THIS WITH THE MIDWIFE, COORDINATING MIDWIFE AND DOCUMENT ON THE ANAESTHETIC RECORD AND ON THE LABOUR WARD WHITEBOARD**
- Give the following 2ml solution made up of: 1ml 0.25% bupivacaine, 15mcg (0.3ml) fentanyl and 0.7ml 0.9% saline (i.e. 2ml 0.125% bupivacaine + 15mcg fentanyl), or 2ml of the low-dose mix (i.e. 2 ml 0.1% bupivacaine and 2mcg/ml fentanyl)
- The likely duration of the resultant analgesia is approximately 30-45 minutes
- Give subsequent top-ups of 2 ml 0.125% bupivacaine (1ml 0.25% bupivacaine + 1 ml normal saline)
- All subsequent top-ups must be given by the anaesthetist
- If the catheter is to be used for an operative procedure, titrate 0.5 ml increments of 0.5% hyperbaric bupivacaine until adequate anaesthesia is achieved
- If a caesarean section is performed with an intrathecal catheter give intrathecal diamorphine 300mcg (0.3ml) at the end of the procedure followed by a 2ml saline flush.
- The catheter must be removed by the anaesthetist after delivery, whilst still on labour ward

Removal of the Tuohy needle and resite of epidural at a different level

- Remove the Tuohy needle and insert another epidural at an adjacent interspace: Seek senior help if epidural placement had been difficult
- Provide epidural analgesia for labour and delivery with the PIEB pump. Due to the potential of the the mix entering the CSF through the puncture and potentially give a higher than expected block, the PIEB should be loaded with 3x 5ml of the mix and started with only the hourly bolus and none of the patient bolus. Once

block height has been established it can then be determined whether it is safe to allow the same patient bolus evey 20mins.

- No change to the management of labour and delivery for the mother, with no restriction on pushing in the second stage
- Communicate the dural puncture with the midwife and document appropriately

If dural puncture is suspected following insertion of the epidural catheter – do not test the epidural catheter in the usual way.

If CSF can be clearly aspirated from the catheter (test for glucose to exclude epidural saline) – leave the catheter in and manage as above for **Insertion of an Intrathecal Catheter**

If dural puncture by the catheter is suspected following an unexpectedly high and rapid block developing after the first epidural local anaesthetic dose, presume the catheter is intrathecal and manage as above.

In cases where it is unclear whether the catheter is epidural, intrathecal or in some other space, remove and resite the epidural as above.

If leaving the epidural catheter intrathecally be aware of the potentially higher risks of infection. If there is doubt regarding the subsequent sterility of the insertion site – remove the catheter and resite if necessary.

Post-Dural Puncture Headache

There is extensive information in the separate guideline on Management of Post-Dural Puncture Headache (PDPH).

This document includes patient information, guidance for the management of PDPH including performance of an Epidural Blood Patch. This guidance and MyCare documentation must be followed in cases of PDPH.

Following an unintentional dural puncture in pregnancy with a Tuohy needle, PDPH develops in over 70% of women.

PDPH is typically most severe across the forehead and worse on sitting / standing. Although most commonly resulting from an unintentional dural puncture, PDPH can also occur after deliberate dural puncture with a spinal needle.

Should unintentional dural puncture occur, the following measures should be taken to try to prevent/minimise development of PDPH:

- Provide analgesia with regular NSAID and paracetamol
- Dihydrocodeine may need to be added but avoidance of constipation is a key element to reducing headache severity
- Keep well hydrated

In addition, an explanation should be given to the mother of what has happened, with reassurance that any headache that develops is likely to go within a few days.

A full and clear record of the findings on Follow-Up Evaluation on MyCare using the post dural headache tab.

Provide the woman with the patient information leaflet for PDPH.

Although PDPH can be extremely distressing for patients, it is normally self-limiting and not associated with serious complications.

Very occasionally, complications may occur including intra-cranial sub-dural haematoma and seizures.

If focal neurological signs or evidence of raised intracranial pressure develop, appropriate imaging must be requested immediately.

In those women who have had a failed EBP then a CT scan should be strongly considered depending also on the strength of the history of a PDPH

Women with a PDPH should be spoken to at least once a day by the obstetric anaesthetist. Full documentation must be made daily on MyCare using the post dural headache tab in the Follow-up Evaluation section.

If a severe headache persists, an epidural blood patch (EBP) should be considered, although it is not usually done before 48 hours because of a lower success rate.

If an EBP is performed please send the post-EBP letter through to the GP. There is a MyCare GP letter proforma that should be appear when noting the EBP procedure.

18.0 Neurological Deficit

Any woman found to have a lower limb neurological deficit postnatally needs the Postnatal Neuro Tab under the Follow-up Evaluation section in MyCare be filled in. Detailed information should be provided including how and when the regional anaesthetic was done and also details of the deficit itself.

This should always be discussed with a consultant. They also require follow up 3 months later in the High Risk Clinic. This can be booked through Orders under Epic. Please specify that this needs to be a face to face (rather than telephone) consultation and when the follow up is required.

References:

*Paech MJ, Doherty DA, Christmas T, Wong CA; Epidural Blood Patch Trial Group. The volume of blood for epidural blood patch in obstetrics: a randomized, blinded clinical trial. Anesth Analg 2011;**113**:126-33*

Turnbull DK, Shepherd DB. Post-dural puncture headache: pathogenesis, prevention and treatment. BJA 2003; 91: 718-29

Sudlow C, Warlow C. Epidural blood patching for prevention and treating post-dural puncture headache. Cochrane Database Syst Rev 2002;2:CD001791

*Zeidan A, Farhat O, Maaliki H, Baraka A. Does postdural puncture headache left untreated lead to subdural hematoma? Case report and review of the literature? Int J Obstet Anesth 2006;**15**:50-8*

19.0 Maternal Collapse and Cardiac Arrest (see appendix)

Although the incidence of cardiac arrest in obstetrics is low 1:30,000, management can be challenging.

Cases of maternal collapse include:

- Haemorrhage
- Thromboembolic disease
  - Pulmonary embolus
  - Amniotic fluid embolus
- Pre-eclampsia/eclampsia
- Anaphylaxis
- Cardiac disease including ischaemic heart disease, congenital heart disease
- Intracranial event including subarachnoid haemorrhage, saggital sinus thrombosis
- Local anaesthetic toxicity
- High regional block (total spinal)
- Drug error (e.g. muscle relaxant, Thio vs antibiotic)

ALS modifications are essential in managing a cardiac arrest in pregnancy and include:

- **Displacement of the uterus to limit aorto-caval obstruction**
  Throughout resuscitation the woman should be tilted/wedged by up to 30°. If however tilting impairs effective CPR, then manage the woman supine with manual uterine displacement.

- **Early delivery of the baby by Caesarean section**
  Perimorbid caesarean section should be performed, with CPR maintained throughout to reduce the impact of the gravid uterus on aortocaval compression and increase the chance of adequate circulation. This should be considered as soon as resuscitation commences and **should be performed after 5 minutes if resuscitation is unsuccessful**.

- **Recognition of the potential obstetric related airway problems and increased risk of regurgitation and aspiration. Early intubation**.

20.0 High Regional Block (Total Spinal)

The signs and symptoms of a total spinal normally develop within 3-4 minutes of a bolus of neuraxial local anaesthetic, with rapid extension of a profound sensory and motor block, respiratory difficulty progressing to apnoea, and significant hypotension and bradycardia.

Unconsciousness follows as the local anaesthetic spreads into the cranial CSF.

Airway support and ventilation with 100% oxygen should be immediately commenced, by facemask initially, with intubation performed as soon as possible using a smaller dose of induction agent.

Management of the cardiovascular system will involve the use of fluids and vasopressors. If severe hypotension and bradycardia consider using adrenaline – dilute 5mg adrenaline in 50ml saline and infuse at 0-20ml/h).

Aortocaval compression must be avoided throughout by lateral tilt or uterine displacement.

Ensure fetal monitoring is maintained throughout. The decision to deliver by caesarean will depend on the individual situation and CTG findings.

The duration of high block following a 'total spinal' is variable, but regression of a very high block is often relatively rapid.

If the correct management is instituted promptly the risk of maternal and neonatal mortality from a 'total spinal' should be very low.

References:

Yentis SM. High regional block: the failed intubation of the new millennium? International Journal of Obstetric Anaesthesia 2001; 10: 159-60

Lucy SJ. Accidental total spinal block. Can J Anaesth 1993; 40: 686-7

Reynolds F, Speedy HM. The subdural space: the third place to go astray. Anaesthesia 1990; 45: 120-3

21.0 Local Anaesthetic Toxicity (see appendix)

Aspirating the epidural catheter before injection will often identify intravascular placement but a negative aspiration test is not wholly reliable.

Whilst problems may occur following the first epidural injection they may present after subsequent top-up.

Adding adrenaline to top-ups may aid recognition of accidental IV injection. Injecting slowly and fractionating the dose given will lessen the peak blood concentration produced, as will early recognition of LA toxicity and stopping the injection.

After injection of a bolus of local anaesthetic, toxicity may develop at any time in the following hour.

CNS signs are often first to present with CVS effects occurring at a higher blood concentration, as outlined below:

Oral numbness Tinnitus/visual disturbance Twitching Convulsions Coma/respiratory arrest Arrhythmia (usually brady-) Cardiac arrest	↓	Increasing blood concentration

Management is dependent on the extent of toxicity, with oxygen and airway support often sufficient during initial stages. **Appendix 5.**

In the event of suspected local anaesthetic toxicity follow the AAGBI guidelines (located in emergency protocol folder in every anaesthetic room)

- **Lipid emulsion is kept on the labour ward resus trolley (labour ward corridor) and in CWH recovery (on the defibrillator trolley)**

22.0 Major Haemorrhage (see appendix)

See Intraoperative cell salvage in obstetrics on the Hub
See Massive obstetric haemorrhage on the Hub

Major haemorrhage in obstetrics can develop quickly and unpredictably.

Haemorrhage may occur ante-partum threatening both mother and fetus, and post-partum.

It is often the post-partum haemorrhage that is associated with greater blood loss and more maternal harm.

Common causes of post-partum haemorrhage can be remembered by the 'four Ts' and include:

Tone – uterine atony
Tissue – retained products
Trauma – genital tract trauma
Thrombin – impaired coagulation

Blood loss may be rapid and surgical intervention is often necessary to stop the bleeding, which for ante-partum haemorrhage is likely to include delivery by caesarean section.

Because of the nature and extent of obstetric haemorrhage, it may be impossible to resuscitate the mother fully before surgery.

If blood transfusion is required urgently, use the O–ve blood kept in the labour ward blood fridge.

Once a major haemorrhage has been identified, the following measures should be taken:

- Instigate the massive haemorrhage protocol (Call 2222 and state – "I wish to activate the massive blood loss protocol" Append
- Get help (ODA, Consultant on call, ITU team as appropriate)
- Large bore IV access (2 if possible)
- Consider early arterial line placement
- Blood warmer
- Patient warming device (Bair Hugger)
- Pressure bags for rapid infusion and if necessary rapid infusion device
- Coagulopathy monitoring and correction using Hemocue and ROTEM to help guide blood and product replacement
- FFP, platelets, tranexamic acid, Calcium as required

A summary of the procedures and interventions that may be required to effectively manage massive PPH are included in the 'major haemorrhage wall-chart' which is on the wall of Theatre 1. See appendix

References:

Walfish M, Neuman A, Wlody D. Maternal haemorrhage. BJA 2009; 103: 147 156

Solomon C, Collis RE, Collins PW. Haemostatic monitoring during postpartum haemorrhage and implications for management. BJA 2012; 109: 851-863

Karlsson O, Jeppsson A, Hellgren M. Major obstetric haemorrhage: monitoring with thromboelastography, laboratory analyses or both? IJOA 2013

Thomas D. Facilities for blood salvage must be available in every obstetric theatre. IJOA 2005; 14: 48-50.

23.0 Pre-Eclampsia

The full management of pre-eclampsia is detailed in the main Maternity guideline Management of Hypertension in Pregnancy but there are several aspects of particular importance to the obstetric anaesthetist, which are outlined here.

Caesarean section

There is little evidence that general or regional anaesthesia improves maternal or neonatal outcome, though whichever technique is selected, major cardiovascular instability should be avoided if possible.

There is good evidence however that regional anaesthesia (either with a spinal or epidural top-up) is unlikely to cause precipitous falls in BP.

Unless contraindicated by low platelets or coagulaopathy, regional anaesthesia should be provided for caesarean section.

General anaesthesia in severe pre-eclampsia may be associated with a number of additional problems including:

- a dangerous surge in BP at laryngoscopy
- difficult airway due to laryngeal oedema

If using a general anaesthetic the pressor response to laryngoscopy must be attenuated at induction (see caesarean section under GA pg 23)

- Alfentanil 10 - 20 mcg/kg (this is probably the easiest to administer)
- Magnesium – 40mg/kg (if not already on magnesium)
- Remifentanil – 1 mcg/kg

Women with severe pre-eclampsia requiring surgery should be discussed with the consultant.

Coagulopathy

Thrombocytopaenia and/or DIC are well-recognised features of pre-eclampsia. If pre-eclampsia is suspected, the platelet count and coagulation status of the woman must be established before neuraxial blockade.

Be alert to a rapidly falling platelet count and ensure blood tests are done every 6 hours in women with severe pre-eclampsia (ideally in severe pre-eclampsia measure platelets and coagulation as close as possible to insertion of a regional block).

Ensure a platelet count >75 $x10^9$/l and INR>1.4 before epidural or spinal injection (or removal of an epidural).

If central venous cannulation is required, be very cautious in the presence of severe pre-eclampsia and possible coagulopathy. The ante-cubital fossa provides a safer route for central venous catheter insertion in the presence of impaired coagulation.

Magnesium Sulphate

Magnesium sulphate is an established part of the management of eclampsia and severe pre-eclampsia.

Blood levels are not routinely monitored during magnesium therapy and the clinical signs of muscle weakness and depressed tendon reflexes are used to assess possible toxicity.

Magnesium has some effects of particular relevance to the anaesthetist, including:

- Potentiation of non-depolarising muscle relaxants, by competing with Ca^{2+} for pre-synaptic influx at the NMJ (suxamethonium is unaffected). If providing a GA to a woman on magnesium, use smaller doses of muscle relaxants and monitor neuromuscular block carefully.

Hypertension

The specific management of hypertension is detailed in the Maternity guideline. It must be remembered that treatment of hypertension does not treat pre-eclampsia but is to protect the mother from cerebrovascular complications.

Fluid management

Despite the presence of hypertension and oedema, in pre-eclampsia the intravascular volume is relatively depleted. After delivery there is mobilisation of ECF into the intravascular space, which reduces the already low colloid osmotic pressure. When combined with an increase in capillary permeability, pulmonary oedema can easily result.

Fluid restriction is typical in management of severe pre-eclampsia and close attention to fluid balance including blood loss is vital.

Post-operative analgesia

Do not give NSAIDs due to risks of renal impairment with pre-eclampsia.

References:

Hull J, Rucklidge M. Severe pre-eclampsia - a case based discussion. Tutorial of the week 2009 no. 158

Henke VG, Bateman BT, Leffert LR. Focused review: spinal anesthesia in severe preeclampsia. Anesth Analg 2013; 117: 686-93

Magpie Trial Collaberative Group "Do women with pre-eclampsia, and their babies, benefit from magnesium sulphate?" The Magpie trial: a randomised placebo – controlled trial. Lancet 2002; 359: 1877-1890

Nelson-Piercy C. Handbook of obstetric medicine. ISIS Medical Media (ISBN 1 899066764)

24.0 Physiological Changes of Pregnancy

During pregnancy, many important maternal physiological changes occur, several of which are highly relevant to anaesthetic practice.

Respiratory

The ribcage is pushed up and out. This may lead to difficulty introducing a laryngoscope into the mouth.

Oxygen consumption and **carbon dioxide production** increase by 15-20% at term.

Airways resistance falls.

Lung volumes change (all values below in litres):

	RV	ERV	FRC	VT	IRV	VC	TLC
Non-pregnant	1.2	1.0	2.2	0.5	2.8	4.3	5.5
Pregnant	1.0	0.8	1.8	0.7	2.8	4.3	5.3

Respiratory rate is largely unchanged, therefore **minute volume** increases by 40%. With the increase in CO_2 production, this results in a fall in arterial PCO_2 of about 20% to approximately 4.2 kPa.

The fall in FRC, combined with the increase in O_2 consumption, results in a far shorter safe apnoeic time before hypoxia develops.

Cardiovascular

Heart rate increases by 20%, **stroke volume** increases by 20%, therefore **cardiac output** increases by about 40%.

Systemic vascular resistance falls significantly, owing to the additional placental circulation plus vasodilatation in the skin, kidneys and liver.

Arterial blood pressure falls slightly in mid-pregnancy (mainly diastolic), but returns to baseline by term.

Venous pressure in the lower part of the body is increased by the presence of an abdominal mass. This produces an increase in the **epidural venous volume** with a resultant decrease in the lumbar epidural and subarachnoid spaces. In the supine and near-supine positions, the abdominal mass can compress both IVC and aorta (**aortocaval compression**) which leads to a marked fall in cardiac output.

The heart is pushed upwards and outwards to produces **left axis deviation** on ECG. A **systolic murmur** related to increased flow is often heard.

Red cell mass increases by 20%, but **plasma volume** increases by 40% and thus **haemoglobin concentration** falls by about 20%.

Gastrointestinal

Gastrin levels are raised, producing an increase in **gastric acid production** and a fall in gastric pH.

The rate of **gastric emptying** is unchanged (contrary to long-held opinion), but the tone of the **gastro-oesophageal sphincter** is reduced leading to a marked increase in the incidence of reflux.

These changes occur from approximately the end of the 1st trimester (14-16 weeks) until about 48 hours post-partum.

Renal

Glomerular filtration rate increases by 40%, though this is matched by a similar increase in sodium and water reabsorption, under the influence of **raised aldosterone levels**. Thus urine volume is largely unchanged. Plasma osmolality and urea and creatinine concentrations all fall, however, with a raised creatinine clearance.

Renin and angiotensin activity is also increased, although the placental vasculature is normally insensitive to the effects of angiotensin II.

The **renal threshold for glucose** falls, and thus glycosuria may occur with a normal blood glucose level.

Ureteric and bladder muscle tone is decreased by the effect of progesterone.

Hormonal

Prolactin levels are increased during pregnancy, and **oxytocin** is increased during labour. Of the other hormones produced by the pituitary, **growth hormone**, **follicle stimulating hormone** and **luteinizing hormone** are all decreased.

Thyroxine levels are raised, but only to match the increased metabolism.

Placental production leads to increased levels of **progesterone**, **oestrogen** and **cortisol**.

Adrenal production of **cortisol** is also increased.

Biochemical

The **albumin** concentration in the mother falls, and is markedly lower than that in the foetus. This leads to accumulation of highly albumin-bound drugs (such as diazepam) in the foetus.

The α_1 **acid glycoprotein** concentration in the mother rises, and thus basic drugs are more highly bound.

Whilst **globulins** are slightly raised, the total colloid osmotic pressure is reduced.

Maternal **alkaline phosphatase** concentration increases 4 fold, the increase resulting mainly from placental production.

The **plasma cholinesterase** concentration falls by about 30%, although the clinical effect of this appears to be negligible.

Anti-thrombin III levels fall, leading to a hyper-coagulable state during pregnancy.

25.0 Spina Bifida Guidelines

There are 2 types of spina bifida - occulta (present in up to 20% of the gen population) and cystica.

Cystica is the more serious form associated with spinal cord tethering.

In patients with a history of spina bifida it is incumbent upon the anaesthetist to understand fully the terminology and extent of the defect before performing neuraxial anaesthesia

Unless the patient has imaging/ report that confirm the absence of spinal cord tethering then it is advisable to avoid regional anaesthesia. In patients with confirmed spina bifida occulta spinal cord tethering is very rare and regional anaesthesia can be performed preferably above the bifid spine (typically L5/S1).

PUBLICATION DETAILS

Author of Clinical Guideline	Obstetric Anaesthetist, Dr Alex Mills
Division/ Department responsible for Clinical Guideline	Specialist Services/CWH/Maternity
Contact details	Ext. 2404, alexmills2@nhs.net
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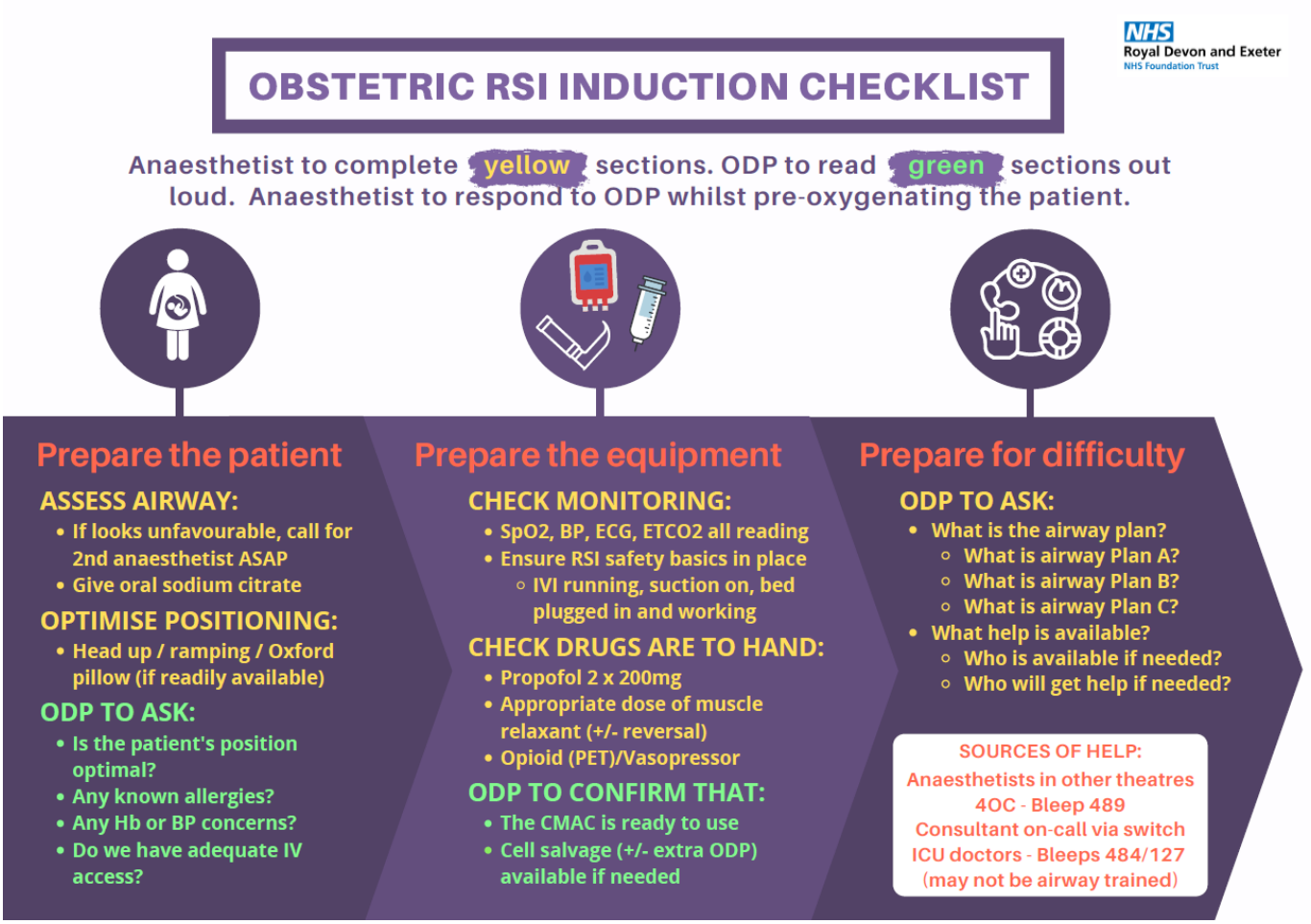
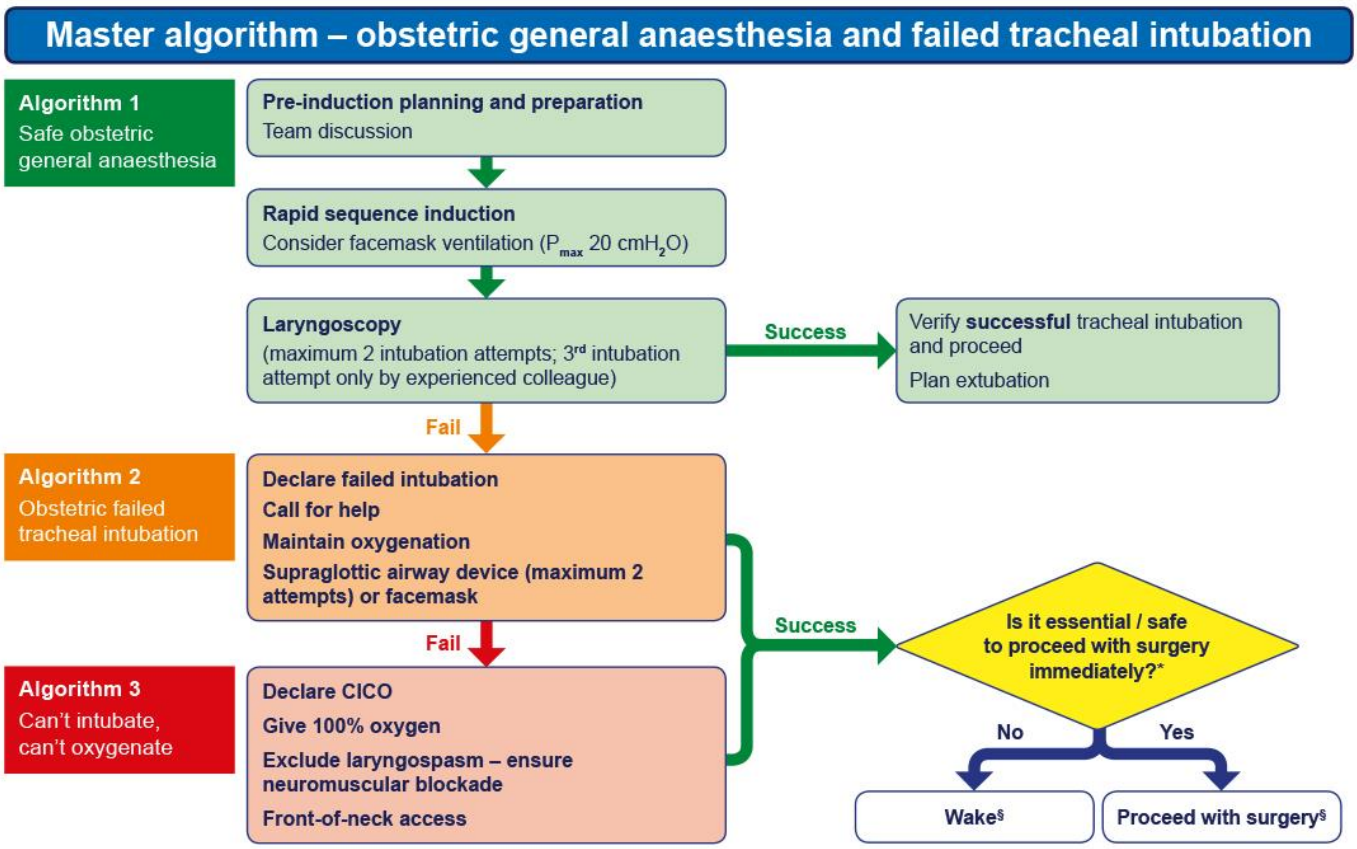
Expiry date	27/04/2025
Date document becomes live	27/04/2022

APPENDIX 1: MORPHINE/ FENTANYL MIX

APPENDIX 2: RSI INDUCTION CHECKLIST



APPENDIX 3: DAS AND OAA FAILED OBSTETRIC INTUBAION GUIDELINES



*See Table 1, §See Table 2

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Algorithm 1– safe obstetric general anaesthesia

Pre-theatre preparation
Airway assessment
Fasting status
Antacid prophylaxis
Intrauterine fetal resuscitation if appropriate

Plan with team
WHO safety checklist / general anaesthetic checklist
Identify senior help, alert if appropriate
Plan equipment for difficult / failed intubation
Plan for / discuss: wake up or proceed with surgery (Table 1)

Rapid sequence induction
Check airway equipment, suction, intravenous access
Optimise position – head up / ramping + left uterine displacement
Pre-oxygenate to $F_{ET}O_2 \geq 0.9$ / consider nasal oxygenation
Cricoid pressure (10 N increasing to 30 N maximum)
Deliver appropriate induction / neuromuscular blocker doses
Consider facemask ventilation (P_{max} 20 cmH_2O)

1st intubation attempt
If poor view of larynx optimise attempt by:
- reducing / removing cricoid pressure
- external laryngeal manipulation
- repositioning head / neck
- using bougie / stylet

Fail

Ventilate with facemask
Communicate with assistant

2nd intubation attempt
Consider:
- alternative laryngoscope
- removing cricoid pressure

3rd Intubation attempt only by experienced colleague

Success

Verify successful tracheal intubation
Proceed with anaesthesia and surgery
Plan extubation

Fail

Follow Algorithm 2 – obstetric failed tracheal intubation

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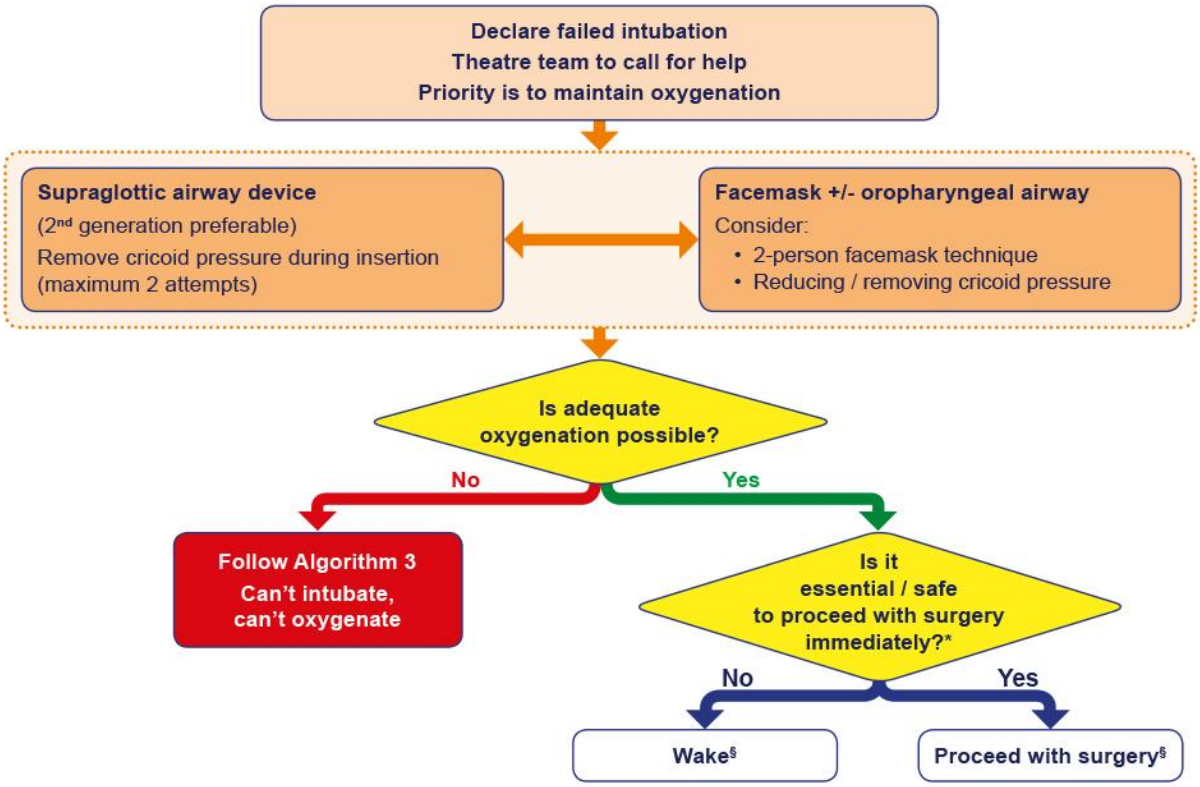
Algorithm 2 – obstetric failed tracheal intubation



*See Table 1, §See Table 2

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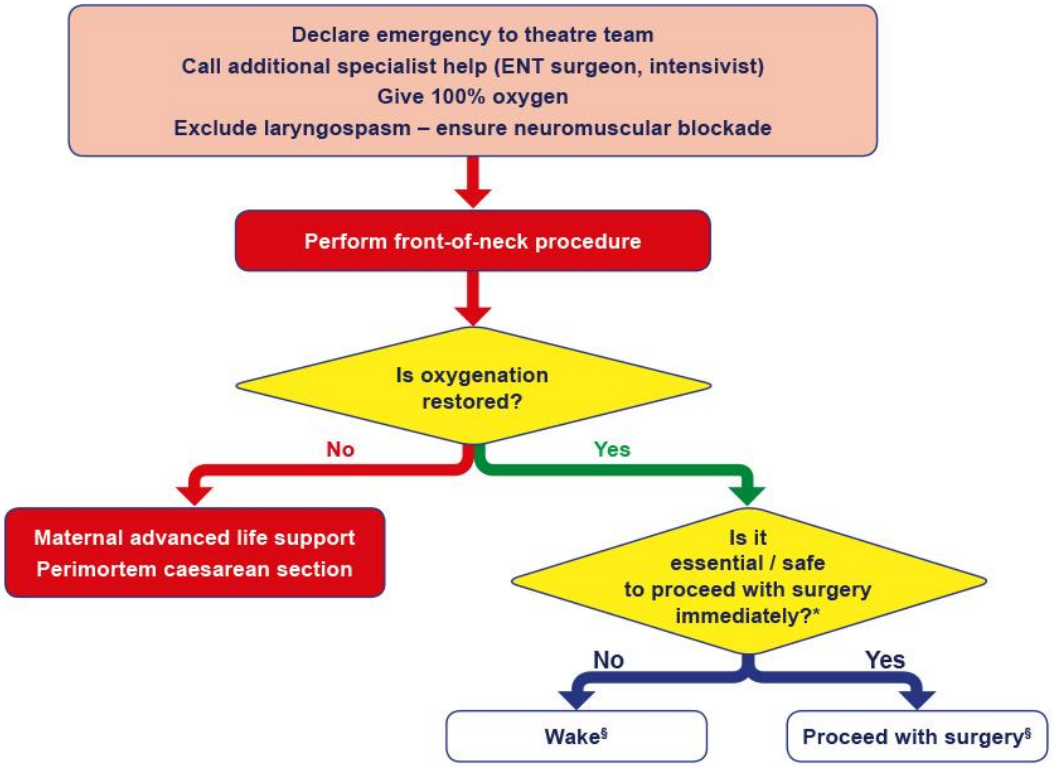
Algorithm 3 – can't intubate, can't oxygenate



*See Table 1, §See Table 2

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Table 1 – proceed with surgery?

	Factors to consider	WAKE ←		→	PROCEED
Before induction	**Maternal condition**	• No compromise	• Mild acute compromise	• Haemorrhage responsive to resuscitation	• Hypovolaemia requiring corrective surgery • Critical cardiac or respiratory compromise, cardiac arrest
	Fetal condition	• No compromise	• Compromise corrected with intrauterine resuscitation, pH < 7.2 but > 7.15	• Continuing fetal heart rate abnormality despite intrauterine resuscitation, pH < 7.15	• Sustained bradycardia • Fetal haemorrhage • Suspected uterine rupture
	Anaesthetist	• Novice	• Junior trainee	• Senior trainee	• Consultant / specialist
	Obesity	• Supermorbid	• Morbid	• Obese	• Normal
	Surgical factors	• Complex surgery or major haemorrhage anticipated	• Multiple uterine scars • Some surgical difficulties expected	• Single uterine scar	• No risk factors
	Aspiration risk	• Recent food	• No recent food • In labour • Opioids given • Antacids not given	• No recent food • In labour • Opioids not given • Antacids given	• Fasted • Not in labour • Antacids given
	Alternative anaesthesia • **regional** • **securing airway awake**	• No anticipated difficulty	• Predicted difficulty	• Relatively contraindicated	• Absolutely contraindicated or has failed • Surgery started
After failed intubation	**Airway device / ventilation**	• Difficult facemask ventilation • Front-of-neck	• Adequate facemask ventilation	• First generation supraglottic airway device	• Second generation supraglottic airway device
	Airway hazards	• Laryngeal oedema • Stridor	• Bleeding • Trauma	• Secretions	• None evident

Criteria to be used in the decision to wake or proceed following failed tracheal intubation. In any individual patient, some factors may suggest waking and others proceeding. The final decision will depend on the anaesthetist's clinical judgement.

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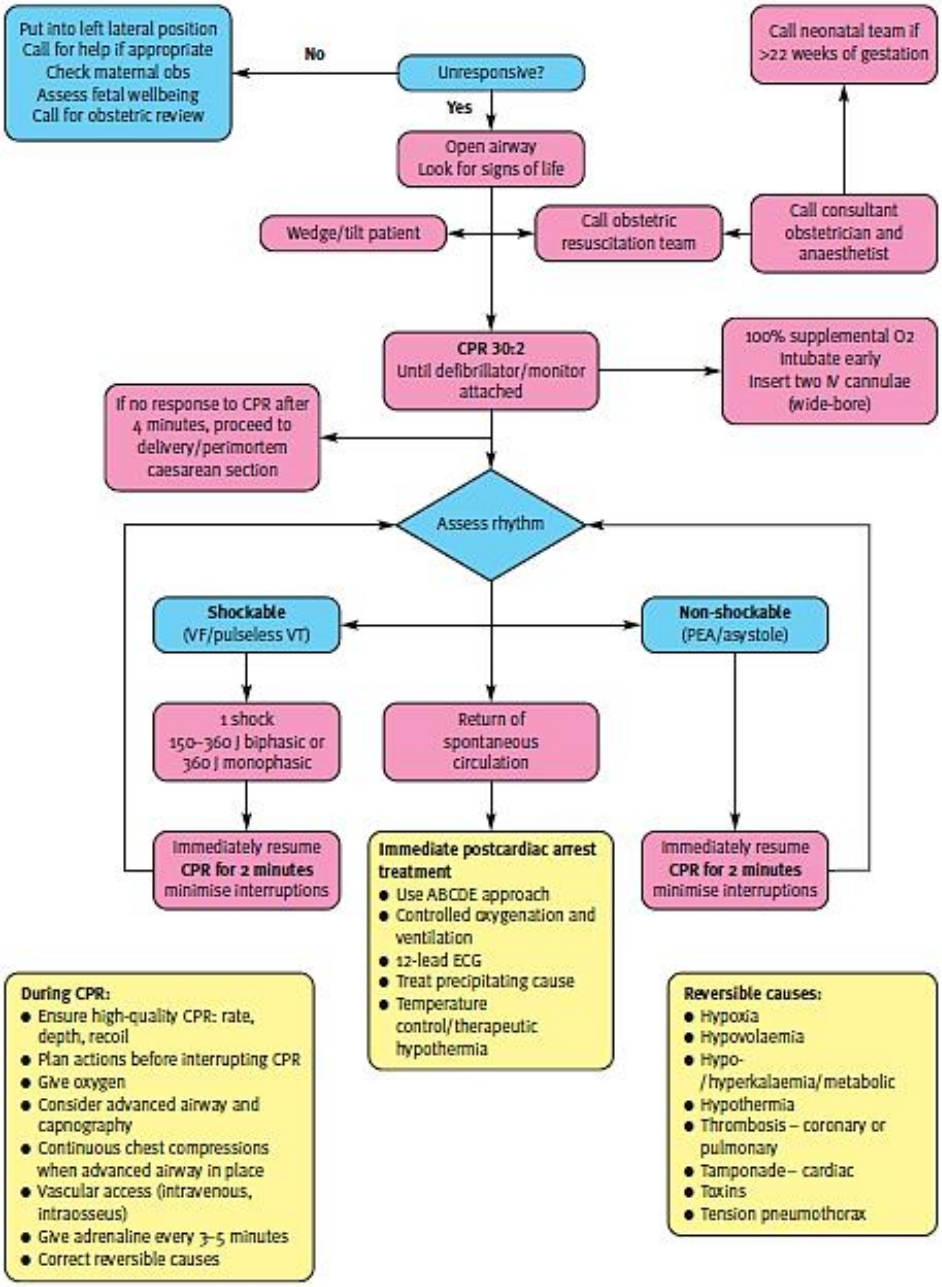
APPENDIX 4: CARDIAC ARREST ALGORITHM

APPENDIX 2 Maternal collapse algorithm



KEY
ABCDE = airway, breathing, circulation, disability, exposure; CPR = cardiopulmonary resuscitation;
RCG = electrocardiogram; PEA = pulseless electrical activity; VF = ventricular fibrillation; VT = ventricular tachycardia

RCOG Green-top Guideline No. 56 22 of 24 © Royal College of Obstetricians and Gynaecologists

APPENDIX 5: MANAGEMENT OF LOCAL ANAESTHETIC TOXICITY

AAGBI Safety Guideline

Management of Severe Local Anaesthetic Toxicity

1 Recognition	**Signs of severe toxicity:** • Sudden alteration in mental status, severe agitation or loss of consciousness, with or without tonic-clonic convulsions • Cardiovascular collapse: sinus bradycardia, conduction blocks, asystole and ventricular tachyarrhythmias may all occur • Local anaesthetic (LA) toxicity may occur some time after an initial injection	
2 Immediate management	• Stop injecting the LA • Call for help • Maintain the airway and, if necessary, secure it with a tracheal tube • Give 100% oxygen and ensure adequate lung ventilation (hyperventilation may help by increasing plasma pH in the presence of metabolic acidosis) • Confirm or establish intravenous access • Control seizures: give a benzodiazepine, thiopental or propofol in small incremental doses • Assess cardiovascular status throughout • Consider drawing blood for analysis, but do not delay definitive treatment to do this	
3 Treatment	**IN CIRCULATORY ARREST** • Start cardiopulmonary resuscitation (CPR) using standard protocols • Manage arrhythmias using the same protocols, recognising that arrhythmias may be very refractory to treatment • Consider the use of cardiopulmonary bypass if available **GIVE INTRAVENOUS LIPID EMULSION** (following the regimen overleaf) • Continue CPR throughout treatment with lipid emulsion • Recovery from LA-induced cardiac arrest may take >1 h • Propofol is not a suitable substitute for lipid emulsion • Lidocaine should not be used as an anti-arrhythmic therapy	**WITHOUT CIRCULATORY ARREST** Use conventional therapies to treat: • hypotension, • bradycardia, • tachyarrhythmia **CONSIDER INTRAVENOUS LIPID EMULSION** (following the regimen overleaf) • Propofol is not a suitable substitute for lipid emulsion • Lidocaine should not be used as an anti-arrhythmic therapy
4 Follow-up	• Arrange safe transfer to a clinical area with appropriate equipment and suitable staff until sustained recovery is achieved • Exclude pancreatitis by regular clinical review, including daily amylase or lipase assays for two days • Report cases as follows: in the United Kingdom to the National Patient Safety Agency (via www.npsa.nhs.uk) in the Republic of Ireland to the Irish Medicines Board (via www.imb.ie) If Lipid has been given, please also report its use to the international registry at www.lipidregistry.org. Details may also be posted at www.lipidrescue.org	

IMMEDIATELY

Give an initial intravenous bolus injection of 20% lipid emulsion 1.5 ml.kg^{-1} over 1 min

AND

Start an intravenous infusion of 20% lipid emulsion at 15 ml.kg^{-1}.h^{-1}

AFTER 5 MIN

Give a maximum of two repeat boluses (same dose) if:

- cardiovascular stability has not been restored or
- an adequate circulation deteriorates

Leave 5 min between boluses

A maximum of three boluses can be given (including the initial bolus)

AND

Continue infusion at same rate, but:

Double the rate to 30 ml.kg^{-1}.h^{-1} at any time after 5 min, if:

- cardiovascular stability has not been restored or
- an adequate circulation deteriorates

Continue infusion until stable and adequate circulation restored or maximum dose of lipid emulsion given

Do not exceed a maximum cumulative dose of 12 ml.kg^{-1}

An approximate dose regimen for a 70-kg patient would be as follows:

IMMEDIATELY

Give an initial intravenous bolus injection of 20% lipid emulsion 100 ml over 1 min

AND

Start an intravenous infusion of 20% lipid emulsion at 1000 ml.h^{-1}

AFTER 5 MIN

Give a maximum of two repeat boluses of 100 ml

AND

Continue infusion at same rate but double rate to 2000 ml.h^{-1} if indicated at any time

Do not exceed a maximum cumulative dose of 840 ml

This AAGBI Safety Guideline was produced by a Working Party that comprised:
Grant Cave, Will Harrop-Griffiths (Chair), Martyn Harvey, Tim Meek, John Picard, Tim Short and Guy Weinberg.

This Safety Guideline is endorsed by the Australian and New Zealand College of Anaesthetists (ANZCA).

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APPENDIX 6: MANAGEMENT OF MAJOR HAEMORRHAGE

DEFINITION:

Minor: 500-1000ml | **Moderate: 1000-2000ml** | **Severe: >2000ml**

BLOOD LOSS IS FREQUENTLY UNDERESTIMATED.

INITIAL MANAGEMENT:

- Call for help**. Switchboard 2222 "Major obstetric haemorrhage"**. Summon: Obstetrician, Anaesthetist, ODA, Midwives, Theatre staff
- **A**irway **B**reathing & **C**irculation
- High flow oxygen via a facemask
- Head down tilt, left lateral if APH
- IV access: 2 x 14 gauge (orange) cannulae
- Take blood: FBC, Coag, 4 Unit X-Match
- Give iv Hartmans via fluid warmer or Rapid infuser
- Use Rotem and Haemacue as guide for blood products
- O-negative blood, if bleeding is uncontrolled
- Inform lab, order blood products: **2467** (blood bank) or Bleep **227**

DIAGNOSIS:

Ante partum haemorrhage (APH)

- Placenta Praevia & Placental Abruption: If severe - deliver baby ASAP.

Post partum haemorrhage (PPH) (> 500ml blood loss post delivery):

- **Tone**: Atonic Uterus (70%)
- **Tissue**: Retained Products (10%)
- **Trauma:** Genital Trauma (20%)
- **Thrombin**: DIC (1%)

SPECIFIC MANAGEMENT:

Specific management of PPH:

Pharmacological:

- Oxytocin (Syntocinon) 5IU slow IV. Repeat once.
- Ergometrine: 500 mcg slowly IV **Caution in PET**
- Oxytocin (Syntocinon): 30 IU in 500mls @ 125 mL/hr IV
- Carboprost 0.25mg IM every 15 min; 8 doses max (0.5mg intramyometrial). **Caution in asthmatics**
- Misoprostol: 1mg per rectum

Surgical:

- Uterine massage if atony
- Bimanual uterine compression
- Ballon tamponade (e.g. Rusch ballon)
- B-Lynch suture
- Ligation of uterine/ internal iliac arteries
- Hysterectomy

Radiological:

Contact radiology consultant via switch: Embolization / arterial balloon occlusion

Blood Loss and Coagulation Management:

- Identify member of team liaise with lab
- Identify member of team to coordinate sample delivery and blood product collection
- Contact consultant haematologist for advice (**139**)
- Give packed red blood cells (RBC)
- After 4U RBC give 1U FFP for each further RBC
- Use ROTEM as guide
- If INR >1.5 give FFP
- Platelets if platelets <$50x10^9$
- Cryoprecipitate or fibrinogen concentrate, if Fibrinogen < 2g/L
- **If DIC suspected fibrinogen concentrate first as instantly available**
- ~~Consider~~ Tranexamic acid 1g IV
- Repeat coagulation studies + regular Hemocue
- Consider 10ml 10% Calcium gluconate

Anaesthetic considerations:

- Call consultant anaesthetist
- Liaise early with ITU: **2424. Bleep 484**
- Avoid hypothermia: early use of Bair Hugger and warmed fluids
- Request cell salvage and ROTEM
- Wash swabs for cell salvage (30% in swabs)
- Monitor urine output and temperature
- Consider arterial line early
- Take regular blood samples for FBC, coagulation, ABG and Hemocue
- If surgery is required, avoid regional anaesthesia if cardiovascular instability: GA and RSI
- Consider CVP line and oesophageal doppler
- Vasopressors may be required despite fluid resuscitation. Use phenylephrine in first instance; Norad (4mg in 40 ml 5% Dex) may be necessary

APPENDIX 7: PRILOCAINE SPINAL

GUIDE: Heavy 2% Prilocaine Spinal

Indications for use:

1. **Simple 3rd Degree Perineal Tears**
2. **Manual Removal of Placentas (MROPs)**
3. **Cervical Sutures**

2% heavy prilocaine will give approximately 75-90 minutes of block duration at a reliable T8-T10. If the procedure is likely to take more time than this then Bupivacaine may be more appropriate. The anaesthetist must be happy to give a prilocaine spinal and a discussion with the obstetrician prior to the procedure should be done to establish the complexity of the case.

Anaesthetists:

- 3mls (60mg) and no opioids for MROPs
- 2mls (40mg) and no opioids for cervical sutures.
- 2mls (40mg) and no opioids for tears. Consider directing bevel downwards and keep patient sitting for up to 5 mins to achieve saddle block.

Obstetricians:

- Consider benefit of local anaesthetic to perineum (discuss dose with anaesthetist)

Appropriate post-operative analgesia on drug chart

Bladder care:

In-out catheter should be encouraged unless there is a clear obstetric reason for a longer period of catheterisation.

If the patient has not been catheterised midwifes should follow the trust clinical guideline: ***Bladder care for women in pregnancy, labour and the postnatal period sections 7, 8 & 9.***

If the patient has been catheterised for obstetric reasons, obstetricians will arrange appropriate time for TWOC

Prilocaine Spinal Information

These spinals work in a similar way to other spinals but they are:

- shorter acting (typically lasting 75-90 minutes)
- may result in a lower block height (usually T8-T10)
- patients do not routinely require catheterisation unless there is an obstetric reason

Which patients may have a Prilocaine spinal:

- **Simple 3rd Degree Perineal Tears**
- **Manual Removal of Placentas (MROPs)**
- **Cervical Sutures**

What does this mean for midwifery post-natal care?

- Bladder care: follow trust clinical guideline: *Bladder care for women in pregnancy, labour and the postnatal period sections 7, 8 & 9*
- Encourage patient to drink fluids postoperatively
- Individualised VTE risk assessment as per guidelines
- Patients will move the legs and be able to mobilise sooner; please encourage early mobilisation
- Analgesia will be prescribed on the drug chart

With surgery in the morning we should aim to discharge patients on the same day unless prevented by obstetric/ paediatric reasons.

The shorter duration of action, fast offset and reduced rate of catheterisation has been shown to improve patient satisfaction.

APPENDIX 8: OAA PATIENT INFORMATION CHART

EPIDURAL INFORMATION CARD
Epidurals in labour – what you need to know

(This card is a summary. Further information is available from www.oaaformothers.info
Please discuss anything that is not clear with your anaesthetist).

Setting up your epidural

- You will need to have an intravenous cannula and maybe a drip.
- While the epidural is being put in, it is important that you keep still and let the anaesthetist know if you are having a contraction.
- Usually takes 20 minutes to set up and 20 minutes to work.
- Some epidurals do not work fully and need to be adjusted or replaced.

Advantages of an epidural

- Usually provides excellent pain relief.
- Sometimes a **spinal** is given first for a quicker effect.
- The dose or type of local anaesthetic can sometimes be altered to allow you to move around the bed. This is a low-dose (or mobile) epidural.
- In general epidurals do not affect your baby.
- Can be topped up for caesarean section if required.

Possible problems with your epidural

- Repeated top-ups with stronger local anaesthetic may cause temporary leg weakness and increase the risk of forceps or ventouse delivery.
- The epidural may slow down the second stage of labour slightly.
- You may develop low blood pressure, itching or a fever during the epidural.
- The epidural site may be tender but usually only for a few days. Backache is NOT caused by epidurals but is common after any pregnancy.

The other side of this card gives important risks of epidurals

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EPIDURAL INFORMATION CARD

Risks of having an epidural or spinal to reduce labour pain

Type of risk	How often does this happen?	How common is it?
Significant drop in blood pressure	One in every 50 women	Occasional
Not working well enough to reduce labour pain so you need to use other ways of lessening the pain	One in every 8 women	Common
Not working well enough for a caesarean section so you need to have a general anaesthetic	One in every 20 women	Sometimes
Severe headache	One in every 100 women (epidural) One in every 500 women (spinal)	Uncommon
Nerve damage (numb patch on a leg or foot, or having a weak leg)	Temporary - one in every 1,000 women	Rare
Effects lasting for more than 6 months	Permanent - one in every 13,000 women	Rare
Epidural abscess (infection)	One in every 50,000 women	Very rare
Meningitis	One in every 100,000 women	Very rare
Epidural haematoma (blood clot)	One in every 170,000 women	Very rare
Accidental unconsciousness	One in every 100,000 women	Very rare
Severe injury, including being paralysed	One in every 250,000 women	Extremely rare

The information available from the published documents does not give accurate figures for all of these risks. The figures shown above are estimates and may be different in different hospitals.

The other side of this card gives information about epidurals for labour pain

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APPENDIX 9: REFERRALS TO HIGH RISK CLINIC

Royal Devon and Exeter NHS
NHS Foundation Trust

High Risk Anaesthetic Pre-assessment Clinic for Obstetrics

Women with anticipated anaesthetic problems or medical disorders should be referred to the Anaesthetic Clinic at an early stage during their pregnancy.
Referrals should be sent to the Anaesthetic Department marked Obstetric Anaesthetic Clinic.

Please send referrals in a timely manner – preferably second trimester.
Ensure Estimated Date of Delivery is on all referrals.

We would expect to be referred the following **antenatally**:

Anaesthetic Issue
Previous problem with anaesthesia – General or Regional
History or family history of malignant hyperthermia
History or family history of suxamethonium apnoea
Women who refuse blood and blood products
Known or anticipated difficult IV access
BMI > 40

Medical Issue
Severe lung disease
Cardiac problems
History of thromboembolism
Bleeding disorder (e.g. VWD, Haemophilia)
Thrombocytopaenia
Hypercoagulability requiring antenatal anticoagulation therapy (e.g. protein S/C)
Haemoglobinopathies (e.g. thalassaemia, sickle cell disease)
Previous neurosurgery (e.g. VP shunt, craniotomy)
Neuromuscular disease (e.g. myasthenia gravis, muscular dystrophy)
Intracranial pathology (e.g. AV malformations, BIH, neoplasm)
History of stroke or intracranial bleeding
Spinal scoliosis/spina bifida
Previous spinal surgery

We would also like to see the following **postnatally**:

Women who had an epidural blood patch
Women with neurology thought to be due to regional anaesthesia
Women who wish to discuss their obstetric anaesthetic experience

All enquires contact the Anaesthetic Department on ext. 2475

High Risk Anaesthetic Pre-assessment Clinic for Obstetrics (version 1.0)
Ratified by Maternity Governance Forum 16/07/2014

APPENDIX 10: ADULT OBSTETRIC AND GYNAECOLOGY ANTIBIOTIC PROPHYLAXIS

Adult Obstetric and Gynaecology Antibiotic Prophylaxis Guidelines

NHS
Royal Devon and Exeter
NHS Foundation Trust

- ✔ **Administer prophylactic intravenous antibiotics 60 minutes or less before the skin is incised (oral and rectal antibiotics must be give one hour prior to the procedure)**
- ✔ **Additional intra-operative doses are required if:**
  - Blood loss is ≥ 1500mls: after fluid replacement repeat initial prophylaxis (all antimicrobials) intra-operatively to maintain therapeutic levels
  - Surgery lasts longer than 4 hours (excluding gentamicin and teicoplanin)
- ✔ **Additional post-operative prophylactic doses, up to 24 hours total, may be given if there is:**
  - (1) gross spillage from a viscus; (2) devitalised viscus; or (3) major break in sterile technique
- ✔ **If an infection is discovered e.g. pus or peritonitis, an appropriate course of antibiotic treatment should be prescribed**
- ✔ **Check MRSA status before prescribing prophylaxis**
  - Treat unscreened patients, and patients with a history of MRSA as positive unless cleared by infection control
- ✔ **Check allergy status**
  - Severe penicillin allergy e.g. angioedema, anaphylaxis or Stevens-Johnson syndrome – avoid all beta-lactams
  - Patients with non-urticarial rash allergy to penicillin may receive cephalosporins in a controlled environment
- ✔ **All antibiotic doses are for adults with normal renal function**
  - Gentamicin doses (based on actual body weight unless obese) should be rounded to the nearest 40mg. If obese, gentamicin dose must be adjusted - see gentamicin dosing calculator
  - Teicoplanin doses (based on actual body weight) should be rounded up to the nearest 200mg

Procedure	1st choice	2nd choice (mild penicillin allergy)	2nd choice (severe penicillin allergy)	MRSA positive (or increased risk)
• Caesarean section • Manual removal of the placenta • Perineal tear involving anus/rectum • Hysterectomy (vaginal, abdominal and laparoscopic) • Laparotomy • Repair of 3rd/4th vaginal tear	Co-amoxiclav 1.2g IV	Cefuroxime 1.5g IV + Metronidazole 500mg IV	Gentamicin 3mg/kg IV + Clindamycin 600mg IV	**Add** Teicoplanin 6mg/kg IV
• Operative vaginal deliveries (Forceps/Ventouse)	Offer prophylactic antibiotics (as above) within 6 hours of birth			
• Diagnostic and operative laparoscopy (where vaginal vault not opened) • Laparoscopic/hysteroscopic sterilisation • Diagnostic/operative hysteroscopy • Endometrial ablation • Endometrial biopsy • Evacuation of retained products of conception (1st trimester)	Antibiotic prophylaxis not routinely recommended			
• Surgical termination of pregnancy	Metronidazole 1g PR at the time of procedure. If Chlamydia test positive **or** no test result but changed partners within 6 months give Doxycycline 100mg BD PO for 7 days			
• Hysterosalpingogram • Laparoscopy with dye hydrotubation	Antibiotic prophylaxis not routinely recommended. If clinically indicated, consider Doxycycline 100mg BD PO for 7 days			

GR-PT-PM02-06 *Approved by the Antimicrobial Stewardship Group: December 2019 Updated: July 2020 Review: July 2022*

APPENDIX 11: OBSTETRIC ANAESTHESIA ESCALATION & ATTENDANCE POLICY

In addition to the resident LW anaesthetist (page 981) covering maternity out of hours (18.00-08.00) a resident senior anaesthetist (4th On Call, page 485) is available from 20.00-08.00. A consultant anaesthetist is accessible 24 hours a day. If the coordinating Midwife or senior members of the Obstetric Team have any specific concerns regarding an obstetric patient, they can contact the 4th On Call via page system or the Consultant On Call via the switchboard. This should be in consultation with the LW anaesthetist.

During the week days there is obstetric anaesthetic cover from 08.00 until 18.00. On the weekend there is 24 hour 4th On Call cover.

The Anaesthetic Department would expect input for challenging or high risk obstetric cases from the 4th on call resident anaesthetist. The 4th On Call should be prepared to help if the labour ward anaesthetist is busy with another patient.

In the following circumstances the input of the 4th On Call should be sought. The list is not exhaustive and judgement should be applied.

- A patient with a postpartum haemorrhage where there is rapid ongoing blood loss and/or a total estimated blood loss of >1500ml
- A patient requiring to return to theatre for further surgical intervention
- A patient with a low, anterior placenta who has had a previous Caesarean section
- A patient with a known or predicted difficult airway
- A patient with significant pre-existing cardiac or respiratory disease – usually an ‘High Risk Anaesthetic Management Plan’ exists for these patients.
- A patient with severe sepsis requiring an anaesthetic for delivery
- A pregnant patient with a non-obstetric pathology that requires an anaesthetic
- The refusal of blood products by a patient in labour or who requires a Caesarean delivery
- A failure of epidural analgesia that has not been resolved by re-siting of the epidural
- If the resident labour ward anaesthetist has any concerns about the management of a pregnant patient e.g. high BMI
- A deteriorating patient where initial management has been unsuccessful

Escalation to the Consultant On Call should be provided if

- The 4th On call is engaged with other activity in the Trust and is unable to attend in a timely manner.
- The 4th On Call requires advice or consultant presence due to either LW acuity or case complexity.

The Consultant Anaesthetist on-call should be informed about the following situations:

- Maternal cardiac arrest or maternal collapse
- A pregnant patient being admitted to the Resuscitation Area of the Emergency Department
- A pregnant patient being admitted to Critical Care
- A patient with an intrauterine death that is complicated by an abruption or requiring delivery by Caesarean section
- A patient with severe pre-eclampsia, HELLP or eclampsia

Attendance of the Consultant Anaesthetist for epidural analgesia

The Anaesthetic Department's expectation is that once an epidural request is made, or an existing epidural fails to provide adequate analgesia, an anaesthetist will attend the patient within 30 minutes of being notified. Please contact the 4th On Call if this is not possible.

Only in exceptional circumstances will an anaesthetist be unable to attend within
60 minutes. If the time scale is over 60 minutes then the On Call Consultant Anaesthetist should be called in to provide this.

Summary of out of hours provision for high acuity on LW or complex case

18.00-20.00 – Resident LW anaesthetist to escalate to on call consultant in main theatres.

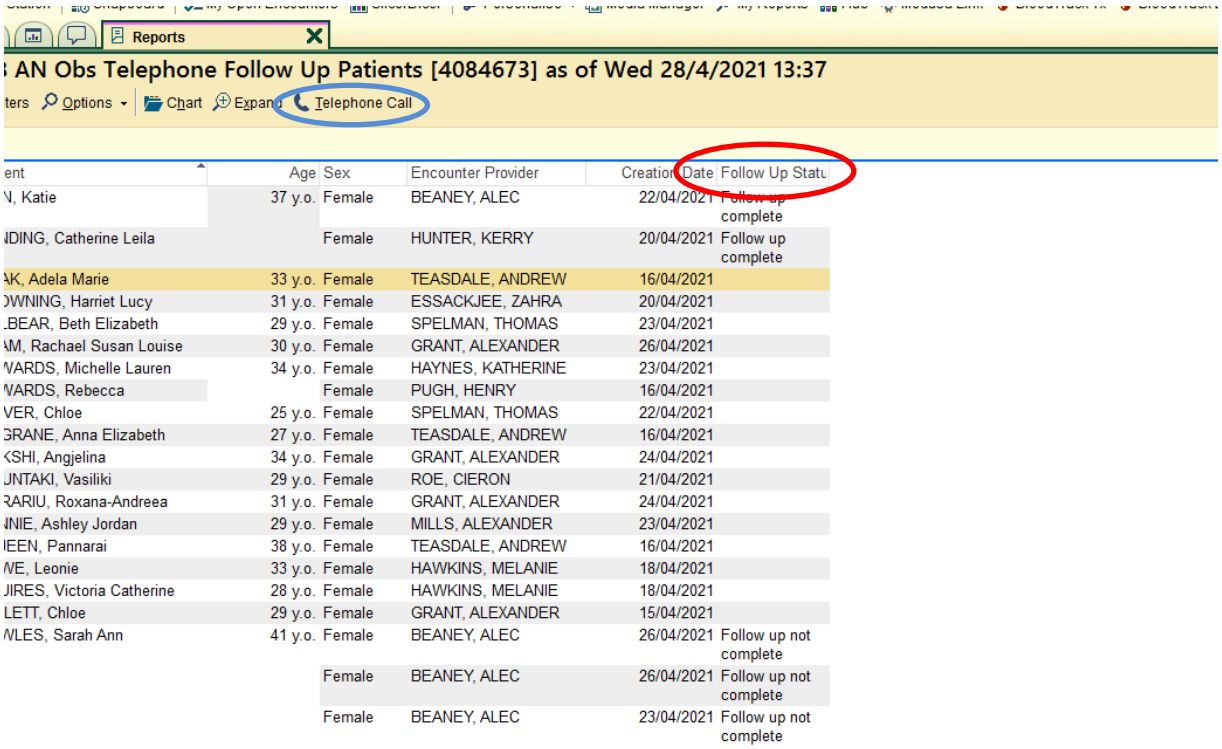
20.00-08.00 – Resident LW anaesthetist to escalate to 4th On Call. If 4th On Call busy then escalate to consultant on call. If complex case then 4th On Call to contact On Call Consultant Anaesthetist.

APPENDIX 12: EPIC TIP SHEETS

Anaesthetic Follow-Up Post Discharge

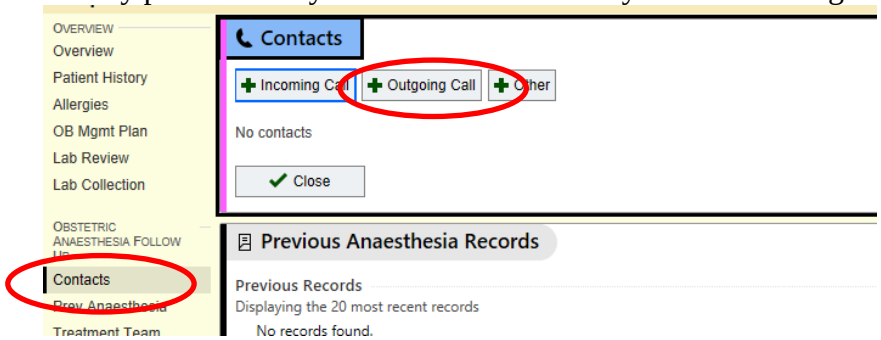
On a daily basis please run the report "AN Obs Telephone Follow Up". It will look like this. Unless a patient has had the 'Follow-up complete' icon clicked as 'Yes' they will be on this report.
The column on the right indicates whether the 'follow up is complete' indicating that we need no further involvement (circled red below). The patient should be automatically removed from the list by the next day. There may well be no patients with this in the column due to the automatic removal.

1) **If it is blank in the right hand column** then the patient left hospital without us seeing them and we need to complete the follow up. Single click on the patient and click telephone call icon (circled blue). *Even though you may not actually make a telephone call*!

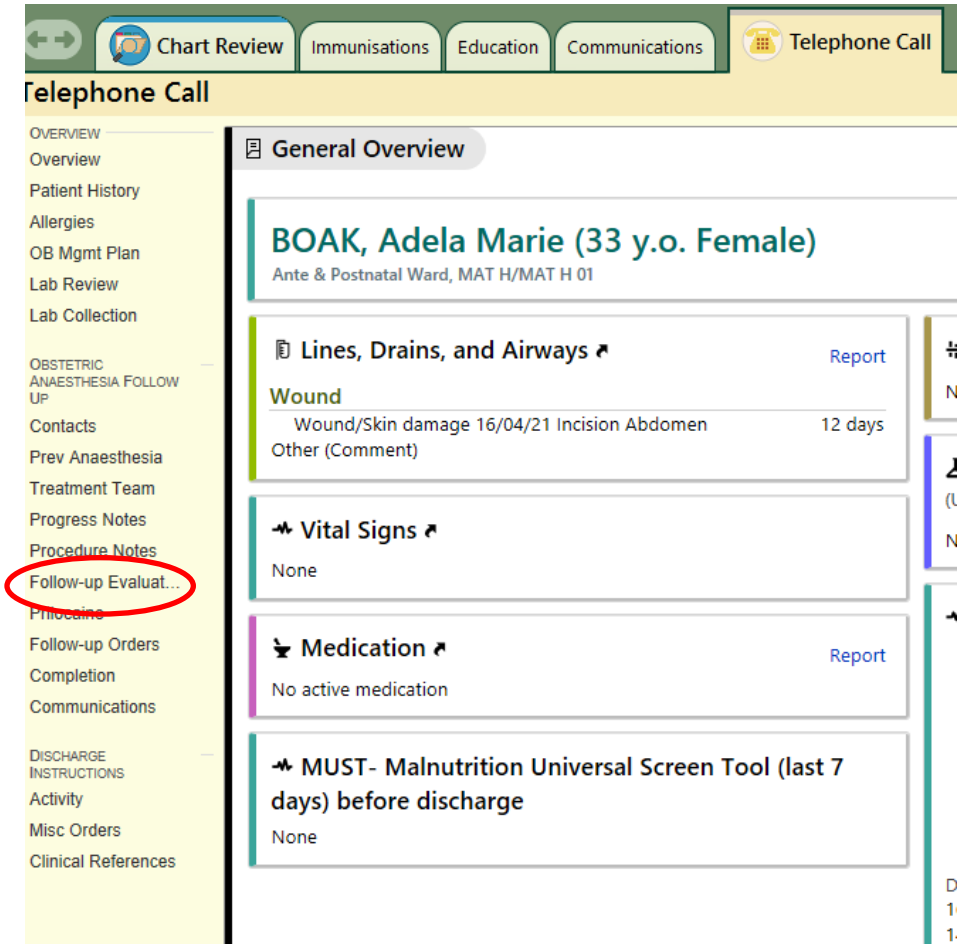


ent	Age	Sex	Encounter Provider	Creation Date	Follow Up Statu
N, Katie	37 y.o.	Female	BEANEY, ALEC	22/04/2021	Follow up complete
DING, Catherine Leila		Female	HUNTER, KERRY	20/04/2021	Follow up complete
AK, Adela Marie	33 y.o.	Female	TEASDALE, ANDREW	16/04/2021	
OWNING, Harriet Lucy	31 y.o.	Female	ESSACKJEE, ZAHRA	20/04/2021	
BEAR, Beth Elizabeth	29 y.o.	Female	SPELMAN, THOMAS	23/04/2021	
AM, Rachael Susan Louise	30 y.o.	Female	GRANT, ALEXANDER	26/04/2021	
VARDS, Michelle Lauren	34 y.o.	Female	HAYNES, KATHERINE	23/04/2021	
VARDS, Rebecca		Female	PUGH, HENRY	16/04/2021	
VER, Chloe	25 y.o.	Female	SPELMAN, THOMAS	22/04/2021	
GRANE, Anna Elizabeth	27 y.o.	Female	TEASDALE, ANDREW	16/04/2021	
KSHI, Angjelina	34 y.o.	Female	GRANT, ALEXANDER	24/04/2021	
UNTAKI, Vasiliki	29 y.o.	Female	ROE, CIERON	21/04/2021	
RARIU, Roxana-Andreea	31 y.o.	Female	GRANT, ALEXANDER	24/04/2021	
INIE, Ashley Jordan	29 y.o.	Female	MILLS, ALEXANDER	23/04/2021	
IEEN, Pannarai	38 y.o.	Female	TEASDALE, ANDREW	16/04/2021	
VE, Leonie	33 y.o.	Female	HAWKINS, MELANIE	18/04/2021	
JIRES, Victoria Catherine	28 y.o.	Female	HAWKINS, MELANIE	18/04/2021	
LETT, Chloe	29 y.o.	Female	GRANT, ALEXANDER	15/04/2021	
VLES, Sarah Ann	41 y.o.	Female	BEANEY, ALEC	26/04/2021	Follow up not complete
		Female	BEANEY, ALEC	26/04/2021	Follow up not complete
		Female	BEANEY, ALEC	23/04/2021	Follow up not complete

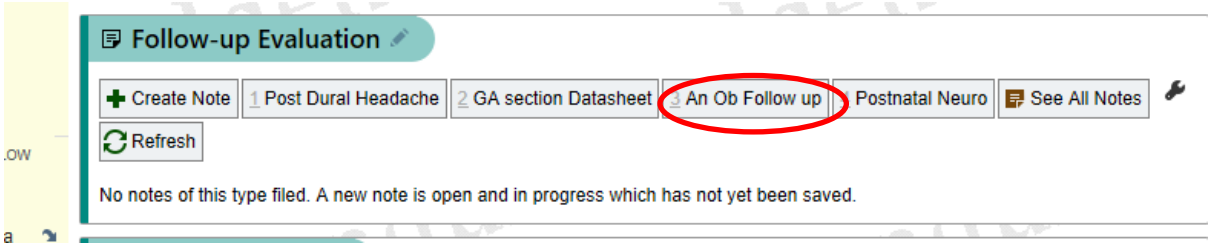
It might be useful to look at the chart review and look at the encounters to see if there have been any obvious postnatal issues reported by the midwifery team. Click the 'Contacts' icon to note that you have made an outgoing call. If there is no answer then write that you left a message to contact maternity triage if there are any problems. If you are satisfied that they can be discharged now (briefly check the midwifery notes) then complete the Follow-Op orders as below.



Then click on Follow-Up Evaluation under the Telephone Call tab.



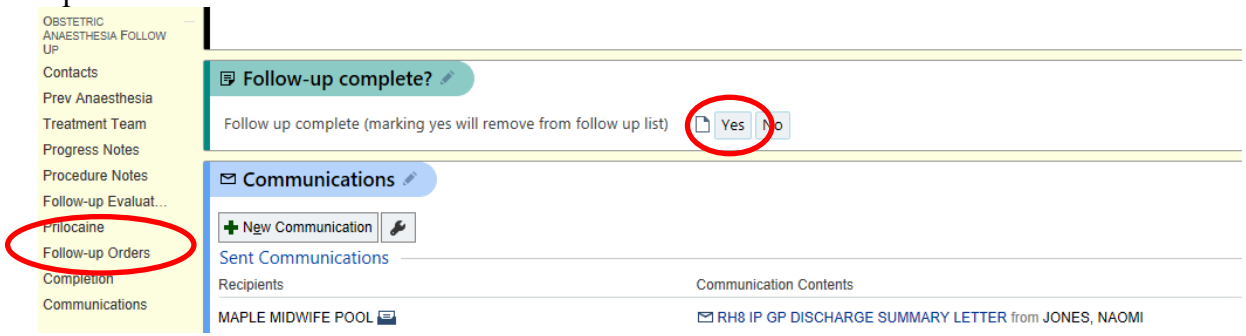
Then click on An Ob Follow Up.



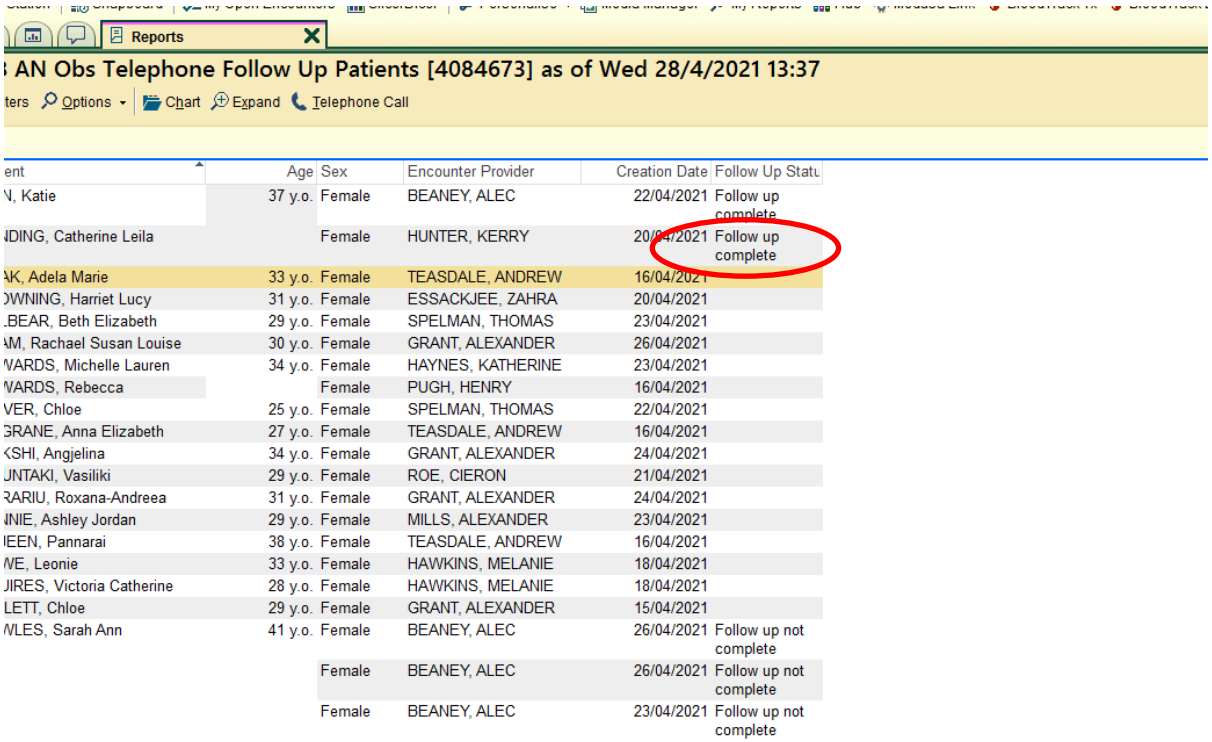
Then click the An Ob Follow up icon on the right side of the screen.

On the left of the screen you can fill in the follow up as you would do an inpatient on the postnatal ward using the click boxes and it does allow for any free text.

Next click the 'Follow-Up Orders' icon then 'Yes' icon under 'Follow up complete'. If the patient has an ongoing issue and you would like them to stay on the report then select 'No'.



Sign and close the encounter and go back to the report page. Refresh the page (top right next to select page). It should now say ‘Follow Up Complete’ and be removed automatically the next day.



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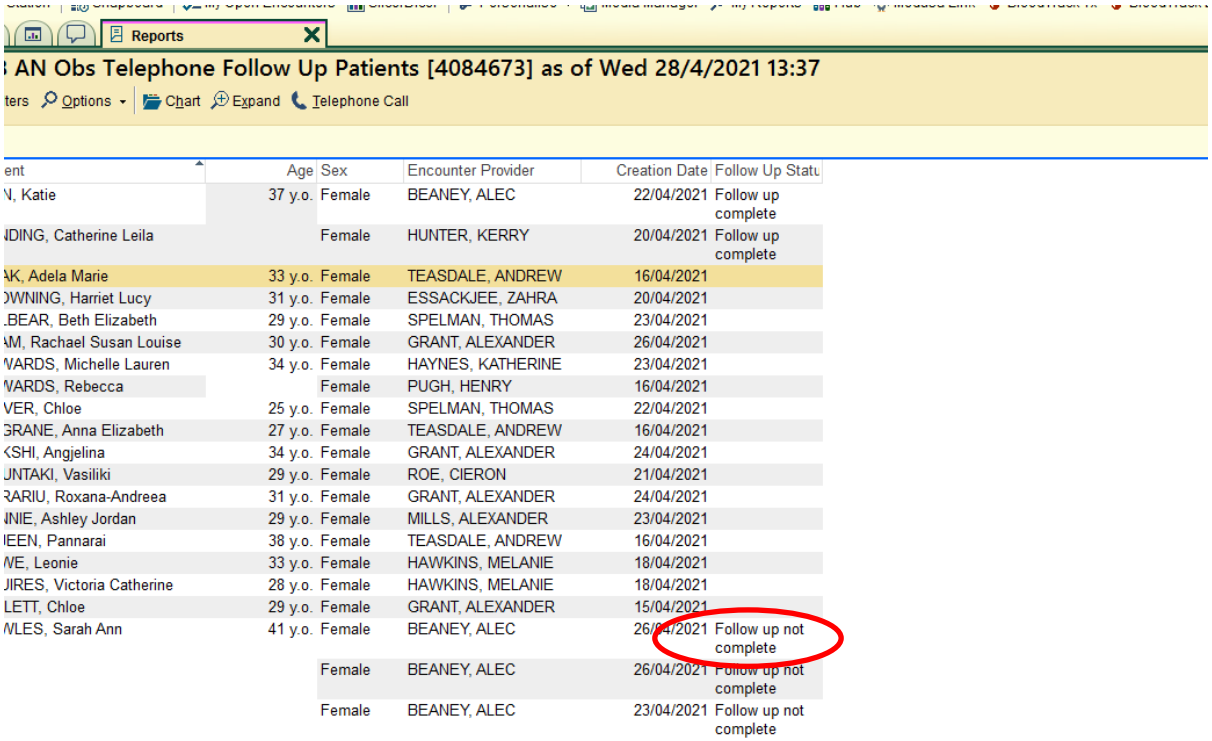
2) **If it says 'follow up is not complete' in the right hand column** that means they are still on our follow up list and we need to contact them. As before, single click the patient, click the telephone encounter. Click on 'contacts' and then 'outgoing call'. Fill this section in.

Then as before click on 'Follow-Up Evaluation, AN Ob Follow Up, An Ob Follow icon' on the right. Fill in this section.
Document in 'Follow-Up Complete' as before. 'Yes' if the patient is fine and the follow up is complete or 'No' if we need to contact them again tomorrow.
They will remain on the list as 'follow up is not complete' (circled red below) until someone clicks 'Yes'.



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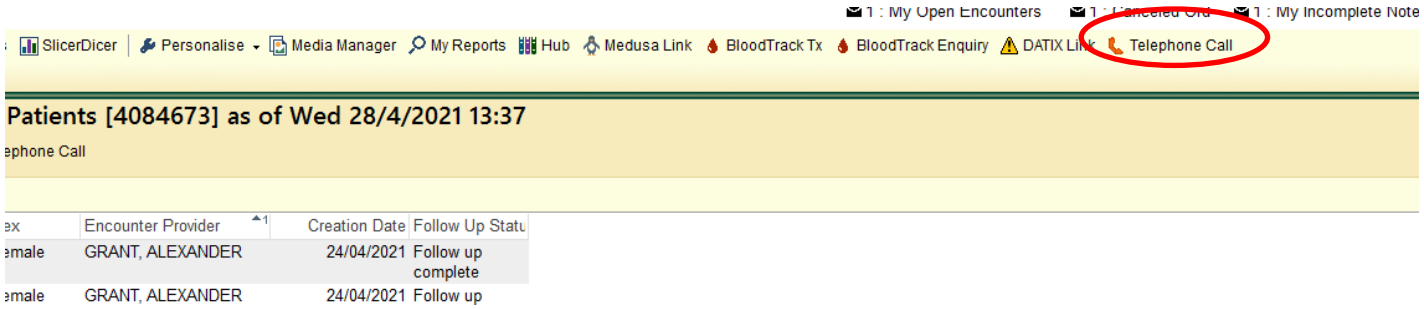
3) **If a patient develops new symptoms and needs to be added to the follow up list**. Click the telephone call icon top right to search for the patient.



This will take you to the familiar screen and go through the same process as for the 'Follow Up Not Complete' patient. This by clicking on the Follow-Up orders icon. Remember to click 'No' on the 'Follow Up Complete' to keep them on the report